Downloaded from gsabulletin.gsapubs.org on October 27, 2010

Geological Society of America Bulletin

Paleomagnetism and oxide mineralogy of Upper Triassic to Lower Jurassic red beds and basalts in the Newark Basin

W. C. McINTOSH, R. B. HARGRAVES and C. L. WEST

Geological Society of America Bulletin 1985;96;463-480
doi: 10.1130/0016-7606(1985)96<463:PAOMOU>2.0.CO;2

Email alerting services click www.gsapubs.org/cgi/alerts to receive free e-mail alerts when new articles cite this article

Subscribe click www.gsapubs.org/subscriptions/ to subscribe to Geological Society of America Bulletin

Permission request click http://www.geosociety.org/pubs/copyrt.htm#gsa to contact GSA

Copyright not claimed on content prepared wholly by U.S. government employees within scope of their employment. Individual scientists are hereby granted permission, without fees or further requests to GSA, to use a single figure, a single table, and/or a brief paragraph of text in subsequent works and to make unlimited copies of items in GSA's journals for noncommercial use in classrooms to further education and science. This file may not be posted to any Web site, but authors may post the abstracts only of their articles on their own or their organization's Web site providing the posting includes a reference to the article's full citation. GSA provides this and other forums for the presentation of diverse opinions and positions by scientists worldwide, regardless of their race, citizenship, gender, religion, or political viewpoint. Opinions presented in this publication do not reflect official positions of the Society.

Notes

Geological Society of America

THE GEOLOGICAL SOCIETY OF AMERICA

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

Paleomagnetism and oxide mineralogy of Upper Triassic to Lower Jurassic red beds and basalts in the Newark Basin

W. C. McINTOSH*
R. B. HARGRAVES
C. L. WEST*

Department of Geological and Geophysical Sciences, Princeton University, Princeton, New Jersey 08544

ABSTRACT

Paleomagnetic and petrographic results from late Carnian to Sinemurian red beds in the Newark Basin suggest that significant components of chemical remanence (CRM) were acquired within a few million years or less after deposition.

The red beds are devoid of magnetite and contain several forms of clearly authigenic hematite. Hematite grains of indisputable detrital origin are rare.

Following removal of Cenozoic components by thermal demagnetization, 56% of the 606 outcrop and subsurface red-bed samples showed normal polarity, 30% were reversed, and 14% were intermediate. The average remanence direction of the normal red beds (D = 7°, I = 26°) was in close agreement with the average direction of the intercalated basalts (D = 7°, I = 27°) and with published Late Triassic/Early Jurassic paleopole data.

The magnetostratigraphic pattern shown by the red-bed and basalt samples is: late Carnian normal, Norian to late Rhaetian mixed polarity, and late Rhaetian to Sinemurian normal. Within the mixed polarity interval is a pattern of thick (as much as 200 m), alternating normal and reversed zones, in some places enclosing numerous thin (0.5 to 20 m), "anomalous" zones of opposite or intermediate polarity. The anomalous zones are associated with lithologic boundaries and are discontinuous along strike; they cannot represent penecontemporaneous records of the geomagnetic field at the time of deposition. Boundaries between the thick polarity zones within the mixed polarity interval, however, are not associated with lithologic boundaries and can be traced along strike for as much as 52 km; these features suggest relatively rapid CRM acquisition.

The normal polarity of the late Rhaetian to Sinemurian red beds agrees well with the normal polarity of correlative basaltic rocks throughout eastern North America and Morocco. Similar correlations can be made with the Norian to late Rhaetian mixed polarity interval. An upper limit of a few million years is indicated for acquisition of CRM sufficient to establish polarity in the red beds; the actual period was probably much shorter.

We conclude that the gross magnetostratigraphic pattern of the Newark Basin strata may be useful for purposes of correlation. The application of the red-bed data to tectonic reconstructions, however, requires caution. The data do not show the expected stratigraphic record of progressive plate movement, and the average remanence direction of the reversed samples (D = 191°, I = −1°) is anomalously shallow.

INTRODUCTION

Controversy exists over the interpretation of the paleomagnetic remanence of red beds (Turner, 1979). Most workers presently agree that the natural remanent magnetization (NRM) of red beds is the vector sum of multiple components of remanence, and that some or all of these components represent chemical remanence (CRM) acquired after deposition. There is strong disagreement, however, as to whether early-formed (primary) components exist and can be isolated and confidently distinguished from late-formed (secondary) components.

On one side of the controversy, researchers believe that some red beds acquired a strong, stable primary component of remanence at or near the time of deposition, either as detrital remanence (DRM) (Elston and Purucker, 1979) or as a CRM carried by authigenic hematite formed within a few thousand years of deposition (Purucker and others, 1980; Baag and Helsley, 1974; Tauxe and others, 1980). These workers believe that such primary components can be isolated by techniques of thermal, chemical, or alternating field demagnetization.

Other workers argue convincingly that the remanence of some or all red beds was acquired through processes long and complex enough to completely obscure any primary components of remanence (Larson and Walker, 1975; Walker and others, 1981; Larson and others, 1982; Larson and Walker, 1982).

An understanding of the mechanism and timing of remanence acquisition in any red bed is critical to assessing the validity of the magnetostratigraphy shown by the unit. If it were possible to actually identify components acquired within time spans which are shorter than single polarity intervals, the magnetostratigraphy of these components will closely resemble the geomagnetic reversal sequence. On the other hand, if the remanence consists of nearly balanced, antiparallel components acquired over two or more polarity intervals, the magnetostratigraphy will be useless for purposes of correlation and dating. The complex nature of chemical remanence acquisition, superimposed on irregularities in the geomagnetic reversal sequence, can produce a magnetostratigraphic record that varies in quality within a single stratigraphic section.

In the present study, the paleomagnetism and oxide mineralogy of the red beds in the Lockatong, Passaic, Feltville, Towaco, and Boonton Formations of the Newark Basin were investigated to determine the magnetostratigraphy and to attempt to evaluate the timing of remanence acquisition. Three factors make these units particularly appropriate for such a study. (1) The red mudstones are, on the average, finer grained and richer in clay than other red beds studied in both paleomagnetic and petrographic detail. In addition, the Newark Basin red beds are valuable for comparative purposes because the rates of diagenetic processes are known to vary between sandstones and mudstones (Blatt and Southerland, 1969; Hubert and Reed, 1978). (2) An independent paleomagnetic record of the geomagnetic field at the time of deposition is provided by the "Watchung" basalts, three multiflow tholeiitic lavas that are intercalated with the red beds in the upper portion of the Newark

*Present addresses: (McIntosh) Department of Geoscience, New Mexico Institute of Mining and Technology, Socorro, New Mexico 87801; (West) Apt. 3, 3303 St. Charles Ave., New Orleans, Louisiana 70115.

Geological Society of America Bulletin, v. 96, p. 463–480, 10 figs., 4 tables, April 1985.

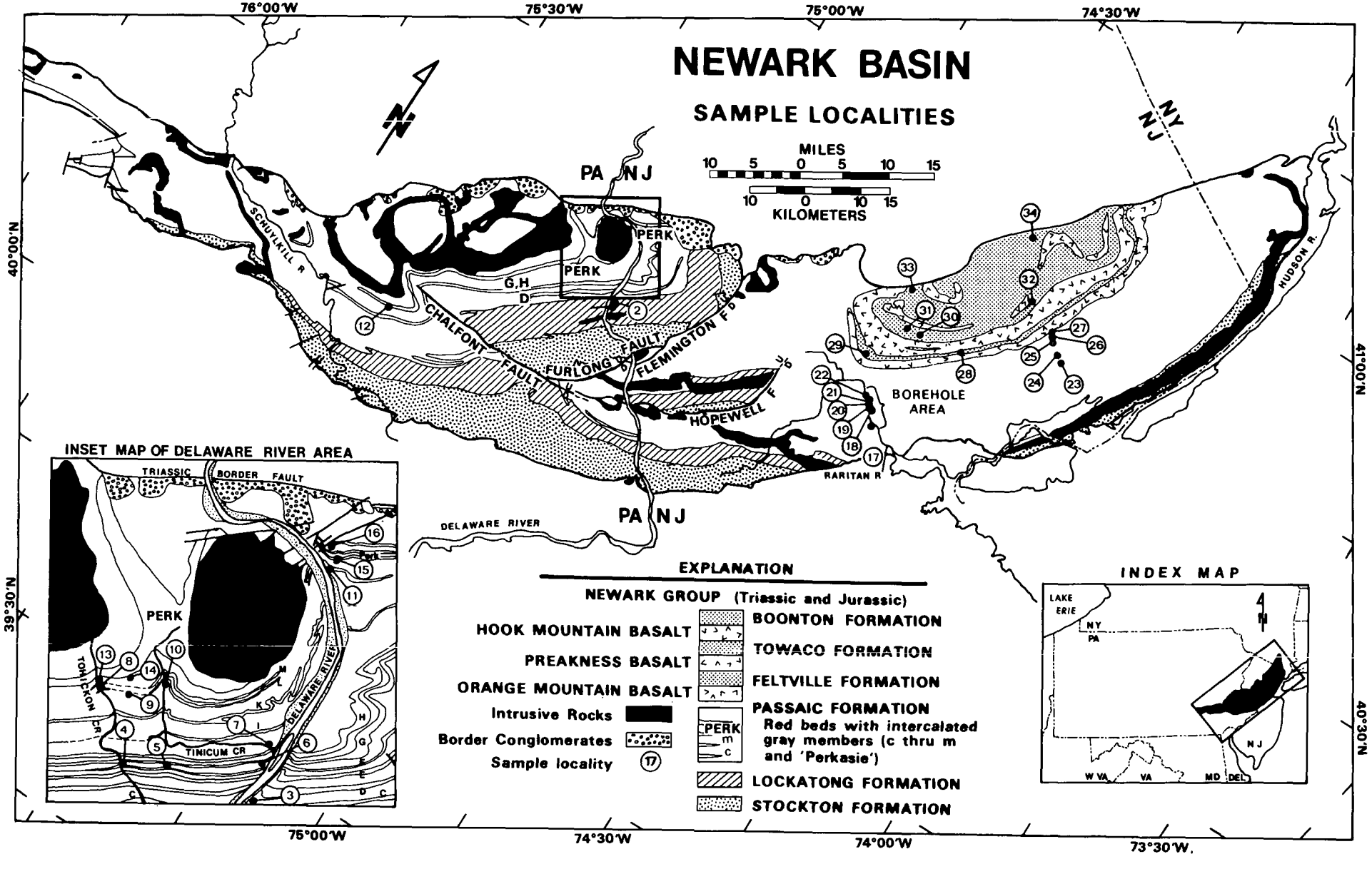


Figure 1. Map showing red-bed paleomagnetic sample localities in the Newark Basin. Subsurface samples were obtained in the area marked "Borehole Area." The Orange Mountain, Preakness, and Hook Mountain Basalts together constitute the "Watchung" basalts.

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

Basin section. (3) Recent biostratigraphic age determinations (Olsen, 1980; Olsen and others, 1982; Cornet and Traverse, 1975) permit detailed correlation of the Newark Basin red beds with the strata in other Triassic/Jurassic basins of eastern North America. Correlations between the Newark Supergroup and Moroccan sections have also recently been advanced (Manspeizer and others, 1978).

Previous paleomagnetic studies have been made of Newark Basin red beds (Du Bois and others, 1957; Bowker, 1960; Opdyke, 1961), but these studies involved few samples of limited stratigraphic distribution and did not employ thermal demagnetization.

GEOLOGIC SETTING

Early Mesozoic sedimentary and volcanic rocks of the Newark Supergroup (Olsen, 1978) are exposed in a series of elongate, southwest-trending structural basins ranging from North Carolina to Nova Scotia. These features were developed as rift valleys during tensional events that were associated with the separation of the African and American plates (Manspeizer and others, 1978). The Newark Basin, 225 km long and as much as 50 km wide, is the largest basin of the series (Fig. 1).

The Newark Basin is a half-graben bounded by normal step-faults along its northwestern edge and broken into a series of blocks by intrabasinal faulting. Within the basin, Late Triassic to Early Jurassic strata rest unconformably on Precambrian and Paleozoic basement rocks and dip 5° to 25° to the northwest. Topographic relief is generally low, and exposures are, for the most part, confined to roadcuts, quarries, and stream valleys.

Deposits within the Newark Basin consist primarily of fine-grained fluvial and lacustrine sediments derived from source areas to the southeast (Van Houten, 1969). Also present are subordinate volumes of (1) coarse-grained alluvial fan deposits near the northwestern border faults, (2) tholeiitic lava flows, and (3) basaltic sills and dikes of Jurassic age (Smith and Noltimier, 1979).

Newark Basin strata have traditionally been divided into three formations termed, in ascending stratigraphic order, the Stockton Arkose (1,800 m thick), the Lockatong Argillite (1,150 m), and the Brunswick Formation (4,800 to 7,800 m) (Kummel, 1898; Van Houten, 1969). Olsen (1978, 1980; see also Olsen and others, 1982) has formally subdivided the sediments and basalts of the Brunswick Formation into a sequence of seven formations: the Passaic Formation, Orange Mountain Basalt (formerly First Watchung basalt), Feltville Formation, Preakness Basalt (Second Watchung), Towaco Formation, Hook Mountain Basalt (Third Watchung), and Boonton Formation. Olsen's terminology is employed in this paper, but the traditional name Watchung basalts is used to refer collectively to the three basalt units.

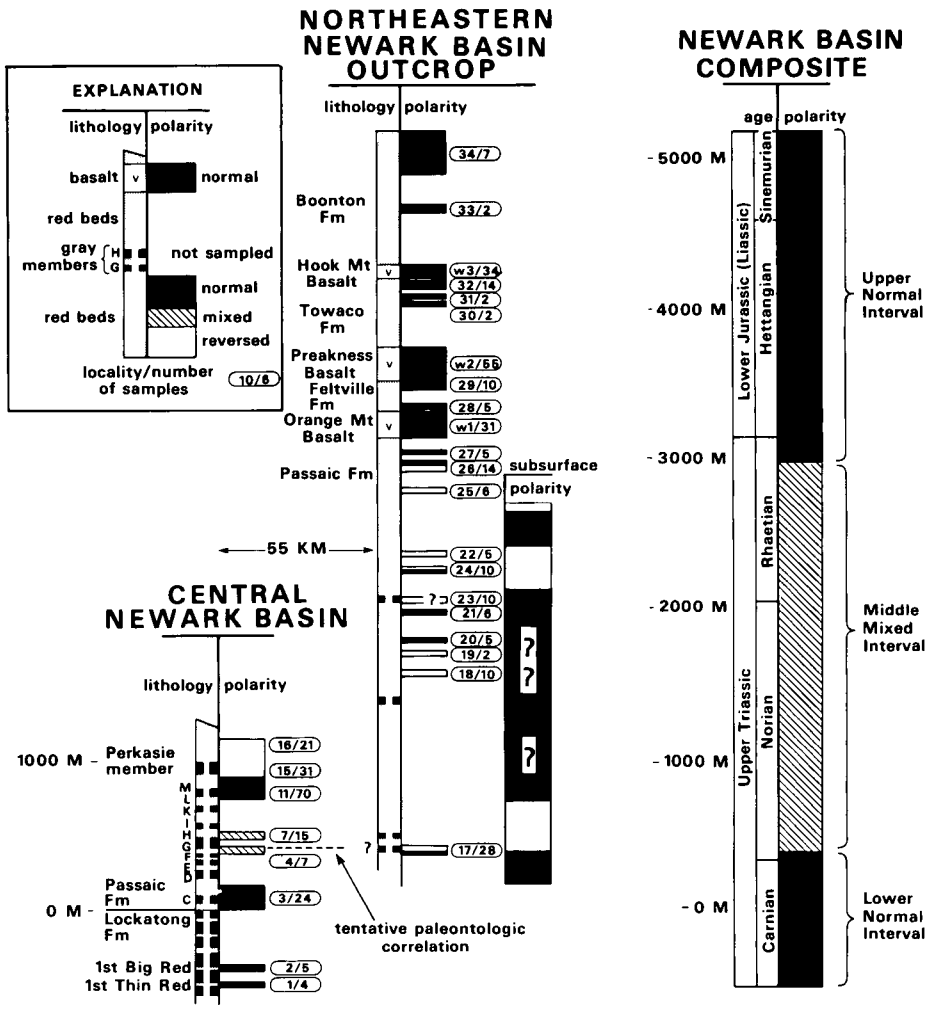


Figure 2. Summary of lithologic sections and magnetostratigraphic data; column at right shows generalized magnetostratigraphy. Ages are based on biostratigraphic data (see Olsen, 1980). Data detailed in Figures 3 and 8 have been generalized here. The sampled thickness at some thinner localities has been exaggerated.

Fossil remains of pollen, spores, pelecypods, arthropods, fish, and reptiles indicate that the Lockatong and Passaic Formations are Upper Triassic (late Carnian to Rhaetian) in age, and the overlying basalts and sedimentary rocks are Lower Jurassic (Hettangian to Sinemurian) (Cornet and Traverse, 1975; Cornet, 1977; Olsen, 1980; Olsen and others, 1982).

Central Newark Basin

The upper 500 m of the Lockatong Formation and the lower 1,200 m of the Passaic Formation are well exposed near the Delaware River in the central Newark Basin. They consist of alternating red fluvial and gray lacustrine members. The red members (4 to 150 m thick) are composed of reddish brown mudstones, shales, siltstones, claystones, and silty sandstones. The gray lacustrine members (0.5 to 87 m thick) consist primarily of shaley to well-indurated claystones and provide excellent stratigraphic control within the otherwise monotonous sequence of red beds. These gray members are relatively uniform in thickness and laterally persistent along strike over distances of at least 60 km. They apparently reflect periodic expansion of alkaline lakes in an area of very low relief (Van Houten, 1962, 1969). McLaughlin (1944, 1946, 1959) mapped the gray members near the Delaware River in detail and gave them informal names (A through M, and Perkasie; Figs. 1 and 2).

Northeastern Newark Basin

The strata of the northeastern Newark Basin area are separated from the central Newark Basin section by two major faults; accurate stratigraphic correlation between the two areas has not been established.

The middle and upper portions of the Passaic Formation are intermittently exposed in the

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

TABLE 1A. CENTRAL NEWARK BASIN SAMPLE LOCALITIES

Locality	n	n_N*	n_R	n_{IN}	n_{IR}	$n_{non\text{-}red}$†	Stratigraphic thickness (m)	Stratigraphic§ position (m)	Location latitude (° ′ ″ N)			Location longitude (° ′ ″ W)		
Lockatong Fm														
1	4	4					8.9	-502	40	26	05	75	04	02
2	5	5					33.4	-395	40	26	24	75	04	02
Passaic Fm														
3	24	22		2			185.1	0	40	27	24	75	04	09
4	7	4	2	1			33.2	388	40	26	33	75	08	50
5	8	4	1	2	1		25.5	395	40	27	18	75	07	26
6	9	6	2	1			18.1	402	40	28	48	75	04	24
7	15	3	3	6	3		24.3	452	40	29	08	75	04	30
8	31	30			1		69.5	800	40	27	54	75	10	26
9	6	6					19.5	815	40	27	44	75	09	27
10	6	6					13.4	837	40	29	00	75	08	01
11	70	70					83.8	745	40	34	16	75	06	21
12	14	4	5	1		4	58.2	883	40	15	20	75	27	39
13	13	6	6	1			49.2	878	40	28	04	75	10	49
14	59	20	25	3	2	9	59.5	867	40	28	17	75	09	50
15	31	5	22		4		188.5	839	40	34	33	75	05	55
16	21		21				102.1	1038	40	34	50	75	06	08

*Criteria for assignment of polarity (N, R, IN, or IR) is discussed in the text.
†Number of non-red samples lacking identifiable Mesozoic remanence.
§Position of base of locality above or below (–) base of Passaic Formation.

TABLE 1B. NORTHEASTERN NEWARK BASIN SAMPLE LOCALITIES

Locality	n	n_N*	n_R	n_{IN}	n_{IR}	$n_{non\text{-}red}$†	Stratigraphic thickness (m)	Stratigraphic§ position (m)	Location latitude (° ′ ″ N)			Location longitude (° ′ ″ W)		
Raritan River–Passaic Fm														
17	28	11	12	2	3		74.1	-2789	40	29	17	74	26	14
18	10	2	7	1			9.7	-1570	40	31	19	74	29	03
19	2		2				6.0	-1437	40	31	35	74	24	37
20	5	4	1				4.1	-1330	40	31	49	74	29	46
21	6	4		2			4.9	-1163	40	32	05	74	30	10
22	5		3		2		4.1	-778	40	32	58	74	31	00
Interstate I-280–Passaic Fm														
23	10		2	4	4		15.3	-1093	40	45	10	74	11	35
24	10	4	4	2			28.7	-394	40	45	30	74	12	28
25	6	1	5				12.6	-359	40	46	19	74	14	20
26	14	4	7	2	1		57.5	-217	40	46	51	74	14	38
27	5	5					12.2	-119	40	47	00	74	14	47
Watchung Syncline														
Feltville Fm														
28	7	5				2	36.5	185	40	40	48	74	23	13
29	10	10					68.4	277	40	36	21	74	32	08
Towaco Fm														
30	2	2		1			3.0	690	40	40	26	74	30	01
31	2	1					1.5	737	40	40	20	74	31	03
32	16	13		1		2	56.7	801	40	48	58	74	19	29
Boonton Fm														
33	3	2				1	36.5	1307	40	44	10	74	33	38
34	7	6		1			284.2	1558	40	53	40	74	23	56

*Same as in Table 1A.
†Same as in Table 1A.
§Position of base of locality above or below (–) base of the Orange Mountain Basalt.

TABLE 2. AVERAGE PALEOMAGNETIC DIRECTIONS OF THERMALLY DEMAGNETIZED RED-BED SAMPLES

Interval	Localities	Biostratigraphic age*	Polarity	n	D	I	K	α_{95}	VPG N lat.	VPG E long.
Lower normal	1 to 3	Late Carnian	N	32	20.6	22.6	16.4	6.5	55.9	67.7
Middle mixed (G/H segment)	4 to 7	Norian	N	27	359.5	53.8	8.4	10.2	83.6	109.4
			R	12	193.6	24.5	5.9	19.5	35.0	89.3
(L/M/Perkasie segment)	8 to 16	Norian	N	126	2.3	20.0	10.9	4.0	59.5	101.1
			R	84	188.0	-6.9	8.7	5.6	52.1	92.5
(Raritan/I-280 segment)	17 to 26	Norian–Late Rhaetian	N	43	6.9	30.2	5.6	10.1	64.8	89.9
			R	47	196.4	3.7	4.0	12.0	44.9	82.1
Upper normal	27 to 34	Late Rhaetian–Sinemurian	N	46	13.4	24.6	7.8	8.1	59.8	73.9
Average of all red beds			N	274	7.0	25.9	7.9	3.2	62.3	90.9
			R	144	190.9	-1.0	5.8	5.4	48.6	89.0
			combined†	418	8.4	17.7	6.2	3.0	57.5	90.0

*Ages after Olsen (1980).
†Calculated with R vectors changed 180°.

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

northeastern Newark Basin. These strata consist of alternating red and gray members that are similar to those of the central Newark Basin, except that the red units here tend to be somewhat coarser and the gray units are thinner (0.5 to 20 m). The maximum thickness of the Passaic Formation in the northeastern Newark Basin is between 3,000 and 6,000 m; estimates are complicated by generally poor exposures and possible hidden strike faults (Kummel, 1898; Van Houten, 1969; Olsen, 1980).

A sequence of six sedimentary and basalt formations overlie the Passaic Formation in the northeastern Newark Basin (Fig. 2). The Feltville Formation (170 m thick), Towaco Formation (345 m), and Boonton Formation (820+ m) consist of poorly exposed red mudstones, siltstones, and sandstones and subordinate non-red sandstones and gray to black siltstones (Olsen, 1980). The intercalated Orange Mountain, Preakness, and Hook Mountain Basalts (the Watchung basalts), average 185 m, 230 m, and 90 m in thickness (Faust, 1975; Manspeizer, 1980), respectively. Each consists of two to three flow units of olivine-poor tholeiitic basalt.

PROCEDURES

Outcrop Samples

Most samples in this study consisted of cores (2.5-cm diameter) drilled in the field and oriented with magnetic and sun compasses with an estimated accuracy of ±3° (McElhinny and Neal, 1967). Where outcrops were fractured or friable, oriented block samples were collected and cored in the laboratory with a net orientation accuracy of ~±5°.

A total of 467 sedimentary samples and 120 basalt samples were collected from outcrops in the central and northeastern Newark Basin. The sedimentary samples included 14 examples of dense gray, indurated claystone and 453 red-colored mudstones, shales, siltstones, and silty sandstones. These samples were taken from 34 localities, each locality being an exposure from which samples were collected at stratigraphic intervals of ~1 to 5 m (Fig. 1; Tables 1A and 1B). These localities ranged over ~5,300 m of late Carnian to Sinemurian section (Fig. 2; Table 2). About 30% of this total thickness is exposed in the sampled outcrops.

Eleven of the 34 localities were chosen to allow assessment of the along-strike extent of magnetostratigraphic features in the central Newark Basin. Immediately beneath each of three persistent, gray, lacustrine members, a set of three to four red-bed localities was selected. The three sets of stratigraphically equivalent localities ranged along strike for distances of 7.5 to 52 km (Figs. 1 and 3; Table 1A). In addition to the sedimentary samples, four to twelve cores were drilled at each of eighteen sites in roadcut exposures of the three Watchung basalts (Table 3).

Raritan Valley Subsurface Samples

Partially oriented subsurface samples of the Passaic Formation were obtained from a series of 3.5- and 5.1-cm-diameter borehole cores as much as 29 m long from the Raritan River valley near Highland Park, New Jersey. The attitude of the strata in the area is quite uniform, and the relative stratigraphic position of each borehole core was established trigonometrically.

The 26 boreholes located in the immediate area of Highland Park provided 77% representation of 272 m of stratigraphic section. The 40 stratigraphically higher borehole cores from farther west along the Raritan River were shorter and more widely spaced, separated by stratigraphic intervals averaging 60 m. The orientation of each borehole core was estimated on the basis of its bedding orientation. Standard oriented samples were then drilled from each borehole core at intervals of ~3 m. By this method, a total of 153 samples was obtained. Sample lithologies ranged from bright red mudstone to reddish brown siltstone. The approximate age of the section represented by the subsurface samples ranges from Norian to Rhaetian (Olsen, 1980).

Laboratory Studies

Stepwise thermal demagnetization of specimens (2.5 cm long) was accomplished in a non-inductively wound furnace within a set of automatically controlled Helmholtz coils which provided a residual field of <35 nano tesla. Alternating field (AF) demagnetization of selected specimens was performed in a dual-axis unit capable of 100 mT (1,000 oe) peak fields and also operated within Helmholtz coils. Remanent vector measurements were made on PAR SM-1 and Schonstedt DSM spinner magnetometers. Remanence directions reported in this paper, except where otherwise noted, have been corrected for geologic dip. Petrographic observations were performed on standard, polished thin sections using a Leitz microscope equipped with vertical-illumination, oil-immersion objectives and capable of effective magnification to 1,250X. Curie balance measurements of induced magnetization versus temperature were performed in air using applied fields of 500 to 1,000 mT.

OXIDE MINERALOGY

Sedimentary Rocks

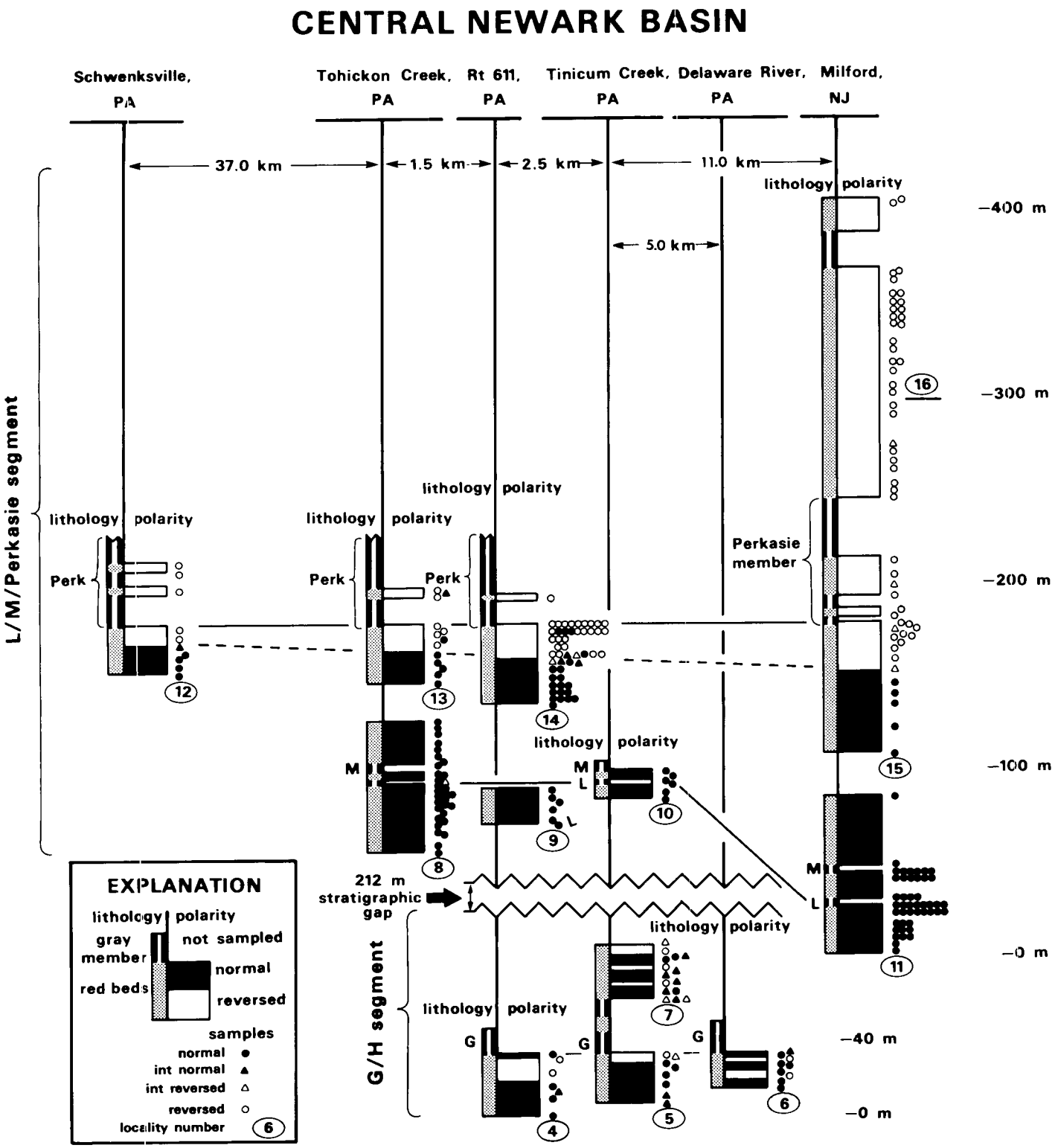
Oxide minerals observed in Newark Basin red-bed samples include translucent, red, microcrystalline hematite, specular hematite, and altered ilmenite. Neither magnetite nor titanomagnetite was seen. Curie temperatures of red-bed samples are near 680 °C, confirming the absence of magnetite.

Microcrystalline hematite is ubiquitous throughout red-bed samples. It is primarily finely disseminated in the clay minerals, but it is also found associated with biotite, within cracks in silicate grains, between overgrowths and detrital grains, and within authigenic minerals. The microcrystalline hematite appears to consist of translucent, disc-shaped crystallites, typically $<1\ \mu$ in diameter but reaching $2\ \mu$ in some cases. The crystallites are present both in masses and as individuals; their reflectivity is low and they appear dull red to yellow in plain light and under crossed polars. An attempt was made to examine the crystallites with a scanning electron microscope (SEM), but it proved impossible to distinguish them from the enclosing clay minerals. The finely disseminated nature of the microcrystalline hematite prevented accurate petrographic estimates of concentration, but it appeared to be the predominant oxide mineral in all of the samples.

The microcrystalline hematite in Newark Basin red beds resembles, petrographically, hematite that was reported from the Triassic Moenkopi Formation of the southwest United States by Walker and others (1981). It is not, however, developed as a void growth mineral, consistent with a general lack of void space observed in the Newark Basin red mudstones. Wherever red color was observed in thin section, microcrystalline hematite was seen to be the staining agent. No evidence was found for the "ultrafine pigment" reported from the Moenkopi Formation by Walker and others (1981).

Specular hematite grains make up ~1% volume of most Newark Basin red-bed samples. They typically range in size from $2\ \mu$ to $20\ \mu$, but some examples reach $100\ \mu$. The following variety of forms of specular hematite was observed. (1) Euhedral to subhedral hexagonal crystals of specular hematite, typically 1 to $15\ \mu$ in diameter, are present both as individuals and as aggregates within the matrix clay of mudstone samples. In addition, yellowish, moderately reflective hexagonal platelets were observed, apparently transitional between individual red, translucent crystallites and crystals of specular hematite. (2) Plates of specular hematite are developed between the cleavage flakes of altered and bleached biotite (the only iron-bearing silicate commonly observed). (3) Stringers and webs of specular hematite are concentrated in the matrix of shale clasts in some samples. (4) Martite, a polycrystalline pseudomorph after magnetite, makes up no more than 2% of the specular grains in most samples. These grains lack ilmenite lamellae and relict magnetite but were identifiable by the distinctive octahedral (111) orientation of individual hematite crystals within the aggregates (Haggerty, 1976).

Downloaded from gsabulletin.gsapubs.org on October 27, 2010



(5) As much as 5% of the specular grains in some red-bed samples exhibit exsolution textures which are diagnostic of formation at magmatic or metamorphic temperatures, that is, (0001) lamellae of ilmenite and lamellae and rodlets of rutile within monocrystalline hematite grains (Haggerty, 1976). (6) About 20% of all specular hematite grains consist of anhedral monocrystalline and polycrystalline grains of uncertain origin. In addition to hematite, some samples contain a few grains of altered ilmenite partially replaced by anatase, leucoxene, and/or rutile.

The preceding types of hematite probably formed by a variety of processes. Microcrystalline hematite is believed to develop, at least in part, by dehydration of iron hydroxide adhering to clay-mineral particles in muddy alluvium (Van Houten, 1968; Larson and Walker, 1975; Turner, 1979). The specular hematite present as euhedral crystals and as alteration products of

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

Figure 3. Data from central Newark Basin Brunswick Formation red-bed samples showing the along-strike extent of magnetostratigraphic features in the Passaic Formation. Note that the polarity boundary at localities 12, 13, 14, and 15 is nearly parallel to the gray lacustrine Perkasie member. Note also that the magnetostratigraphy at localities 4, 5, 6, and 7 is not continuous along strike and is more complex than that of the other sites in this figure. [Some data points (all normal) are omitted from locality 11.] The criteria for assignment of sample polarity are discussed in the text.

TABLE 3. PALEOMAGNETIC DATA FROM THE "WATCHUNG" BASALTS

Unit	Site	N samples	n spec	20 mT D	I	K	σ_{95}	Location (° ′ ″ N)			(° ′ ″ W)		
Orange Mountain	1	8	16	354.5	25.6	39.7	8.9	40	54	30	74	10	45
("First Watchung")	2	5	10	4.0	28.4	30.0	14.2	40	47	45	74	15	32
Basalt	3	6	6	0.2	31.4	108.4	6.5		47			15	
	4	8	8	355.1	32.6	350.3	3.0	40	47	51	74	15	32
	5	4	4	5.3	30.9	218.4	6.2	40	38	37	74	37	55
	mean	31	44	359.8	29.9	251.4	4.8						
Preakness	6	4	8	14.4	0.8	382.8	6.3	40	47	28	74	14	57
("Second Watchung")	7	13	13	15.8	9.9	39.7	6.7	40	48	01	74	16	36
Basalt	8	6	12	25.2	31.1	337.0	3.7		"			"	
	9	10	10	11.2	17.3	272.8	2.9	40	38	39	74	36	03
	10	6	6	26.5	32.2	17.8	16.3	40	38	34	74	36	38
	11	4	4	9.4	22.6	401.8	4.6	40	40	12	74	37	51
	12	4	4	6.5	49.2	229.7	6.1	40	40	16	74	37	14
	13	4	4	8.4	40.3	357.1	4.9	40	40	21	74	36	47
	14	4	4	3.8	45.8	98.7	9.3	40	40	38	74	36	29
	mean	55	65	13.9	27.9	21.1	11.5						
Hook Mountain	15	10	20	7.1	29.5	131.1	4.2	40	51	43	74	20	29
("Third Watchung")	16	4	9	4.8	12.2	152.0	4.9		"			"	
Basalt	17	15	15	357.6	36.2	59.9	5.0	40	48	58	74	19	29
	18	5	5	5.9	30.1	169.9	5.9	40	46	07	74	29	31
	mean	34	49	4.0	27.1	54.9	12.5						
	Basalt mean												
	sites	18		7.7	28.4	31.8	6.2						
	samples	120	158	7.2	27.2	25.8	2.3						

TABLE 4. TRIASSIC-JURASSIC PALEOMAGNETIC POLE-POSITIONS FOR NORTH AMERICA

Unit	Age	N	k	σ_{95}	d_m	d_p	N lat.	E long.	Source
Chinle Fm.	TR_u (Carnian)	191	45	10.1	10.1	5.1	58.2	76.1	Reeve and Helsley (1972)
Newark Basin									
red beds (normal)	TR_u–J_l (Carnian to Sinemurian)	274	8	3.2			62.3	90.9	this study
(reversed)	TR_u (Norian to Rhetian)	144	6	5.4			48.6	89.0	this study
Watchung basalts	J_l (Hettangian to Sinemurian)	120	26	2.3			63.0	90.1	this study
Turners Falls sandstone, Connecticut	J_l (Hettangian)	61		7.8			68.2	79.6	Brown (1979)**
Newark Trend igneous, Group 1	J_l (Hettangian to Sinemurian; ~190 Ma)	72*	56	2.3			63.0	83.2	Smith and Noltimier (1979)
Kayenta Fm., Utah	J_l (Pleinsbachian)	105	32	2.5	2.5	1.4	61.7	72.7	Steiner and Helsley (1974)
Newark Trend igneous, Group 2	J_l (Toarcian; ~175 Ma)	156*	92	1.4			65.3	103.2	Smith and Noltimier (1979)
Somerville Fm., Utah	J_m (Callovian)	15	71	7.6	5.4	3.2	67.5	110.6	Steiner (1978)

N is number of samples, or, where starred, number of sites; ** indicates VGP calculated from published remanence directions.

biotite and shale clasts is also authigenic. The origin of martite is debated; analyses of modern, sand-sized sediments indicate that 10% to 40% of the volume of magnetite grains may alter to hematite in the source area or during transport (Van Houten, 1968; Elston and Purucker, 1979; Walker and others, 1981). Finally, the hematite with high-temperature exsolution textures is certainly of detrital origin. This might be true of some or all of the undistinguished anhedral grains, as well.

Polished sections of gray, lacustrine claystones were found to be altogether devoid of hematite. Amorphous iron hydroxides were observed within euhedral voids representing casts of dissolved pyrite crystals.

Basalts

Euhedral to subhedral intergrowths of magnetite and ilmenite, 40 μ to 2 mm in diameter,

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

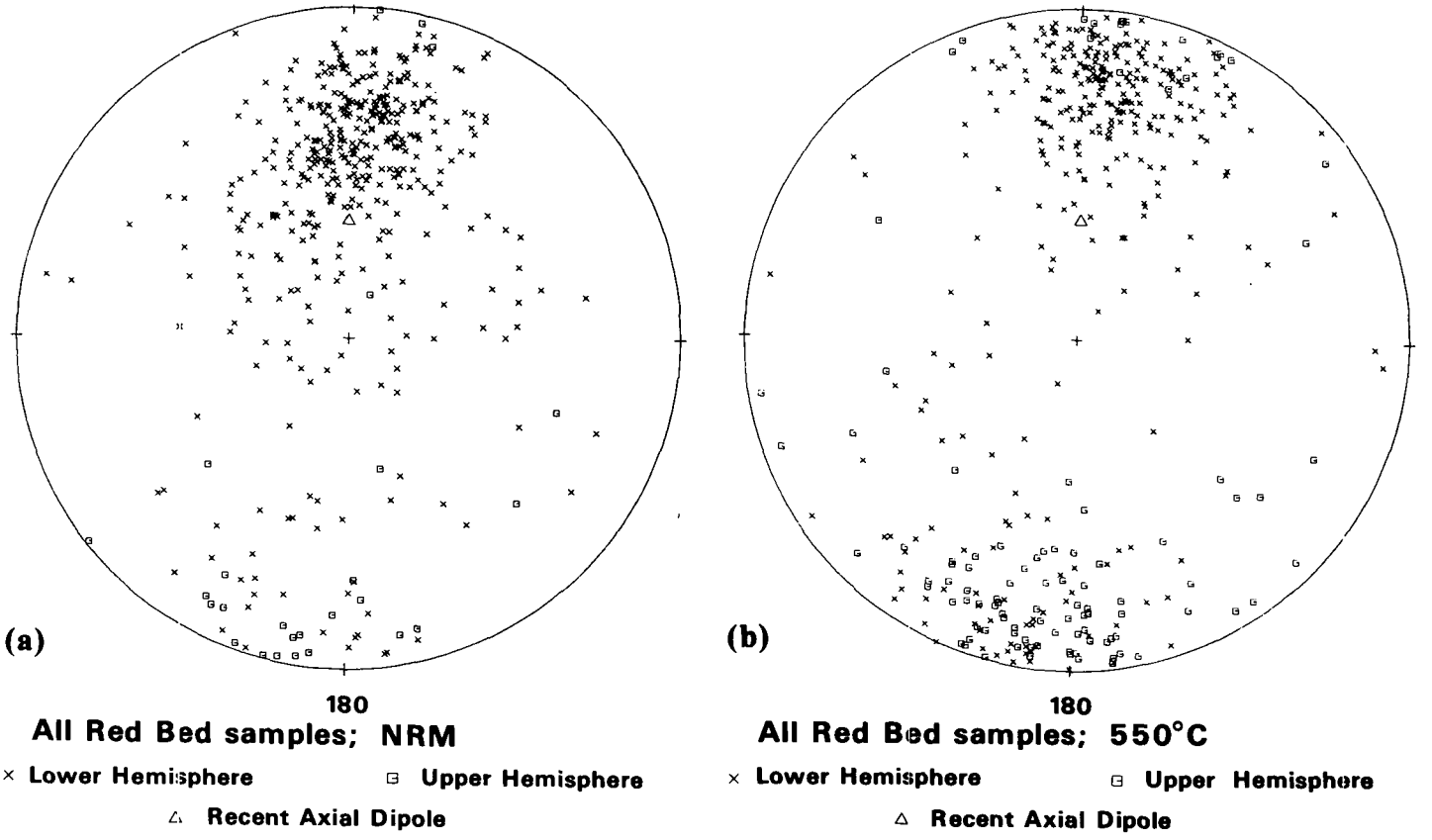
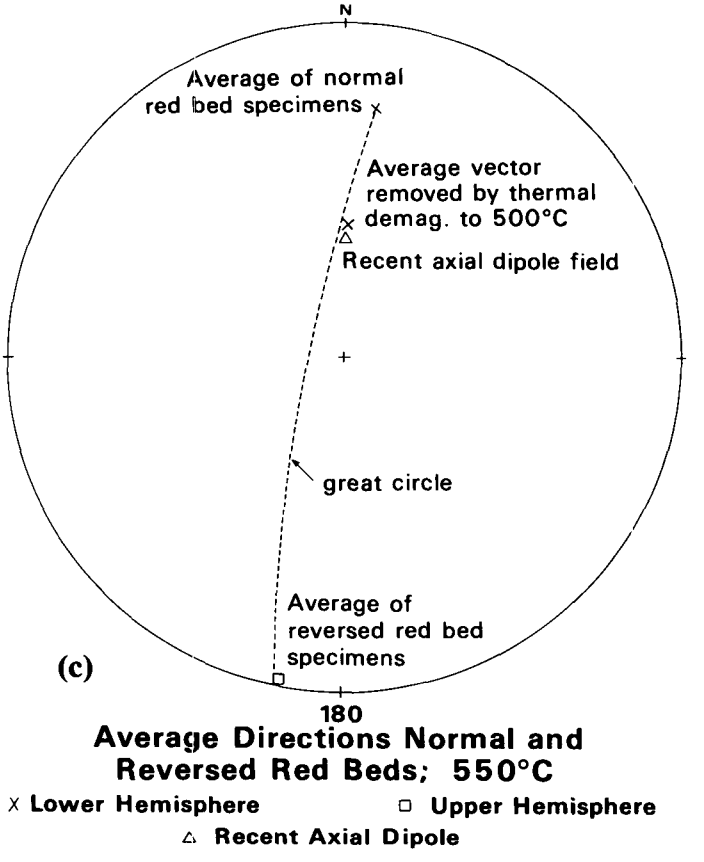


Figure 4. Equal-area projections of remanent vectors of all red-bed samples studied. (a) = NRM. (b) = 550 °C; demagnetization produces nearly antipodal clusters of normal and reversed sample directions. (c) = average of demagnetized N and R and relation to recent axial dipole direction.

were observed in five polished thin sections from finely crystalline margins and coarsely crystalline interiors of the Orange Mountain and Preakness Basalts (First and Second Watchung basalts). Ilmenite commonly makes up 30% to 40% of each oxide grain and consists of either 5 to 40-μ-thick lamellae developed preferentially along a single crystallographic direction or elongate, subhedral patches as much as 500 μ long. Euhedral to subhedral patches of magnetite, as much as 800 μ in maximum dimension, make up the remainder of each oxide grain. Most of the grains exhibit a slight to moderate degree of marginal granulation or dissolution, which commonly affects the magnetite more than the ilmenite. Minor amounts of specular or microcrystalline hematite are associated with cracks, voids, or margins of some magnetite-ilmenite grains. Curie temperatures of basalt samples range from 550 to 580 °C, indicative of low-titanium magnetite.

The magnetite-ilmenite grains in these basalts represent moderate, high-temperature deuteric



Downloaded from gsabulletin.gsapubs.org on October 27, 2010

oxidation of originally homogeneous titanomagnetite (oxidation stage C3 of Haggerty, 1976). The coarse nature of these intergrowths is related to slow cooling of the thick lava flows (in some cases >100 m thick) which constitute the Watchung basalts. The granulation or dissolution of the opaque grains is probably a moderate-temperature deuteric or hydrothermal effect (Ade-Hall and others, 1968). The small amount of hematite and lack of maghemite in the basalt samples indicate that the degree of low-temperature oxidation is slight.

The above observations suggest that the remanence carried by the Watchung basalts is a thermoremanent magnetization (TRM) without significant CRM components. The intergrowths of magnetite are large enough to carry multidomain remanence.

PALEOMAGNETISM OF THE SEDIMENTARY ROCKS

Demagnetization Behavior

Red Outcrop Samples. NRM measurements of specimens of the 453 red samples showed intensities ranging from 9×10^{-2} to 8×10^{-4} A/m and averaging 7×10^{-3} A/m (A/m = 10^3 emu/cm^3). No significant correlation between remanence intensity and lithology of the samples was observed. NRM directions showed a high degree of scatter (Fig. 4a).

Specimens of red outcrop samples were thermally demagnetized stepwise to maximum temperatures of 550 to 720 °C. Orthogonal plots (Zijderveld, 1967) of the demagnetization behavior of representative specimens are shown in Figure 5.

Most of the red specimens reached stable endpoints between 450 and 650 °C. More than half of the specimens showed normal directions, small directional changes, and steadily declining intensities (Figs. 5a and 5b). Other specimens showed generally reversed directions following small (Fig. 5c) to large (Fig. 5d) directional changes. The intensities of 90% of these reversed specimens increased at some stage of thermal demagnetization, indicating removal of normal components. The remaining minority of specimens exhibited intermediate directions and generally moderate intensities during thermal demagnetization (Figs. 5e and 5f).

Thermal demagnetization of red specimens suggests a maximum effective blocking temperature of 630 to 680 °C; at higher temperatures, most specimens exhibited very low intensities ($<10^{-4}$ A/m) and erratic intensity or directional behavior. This behavior may reflect the acquisition of TRM due to imperfections in laboratory field cancellation (±35 γ). No clear evidence of

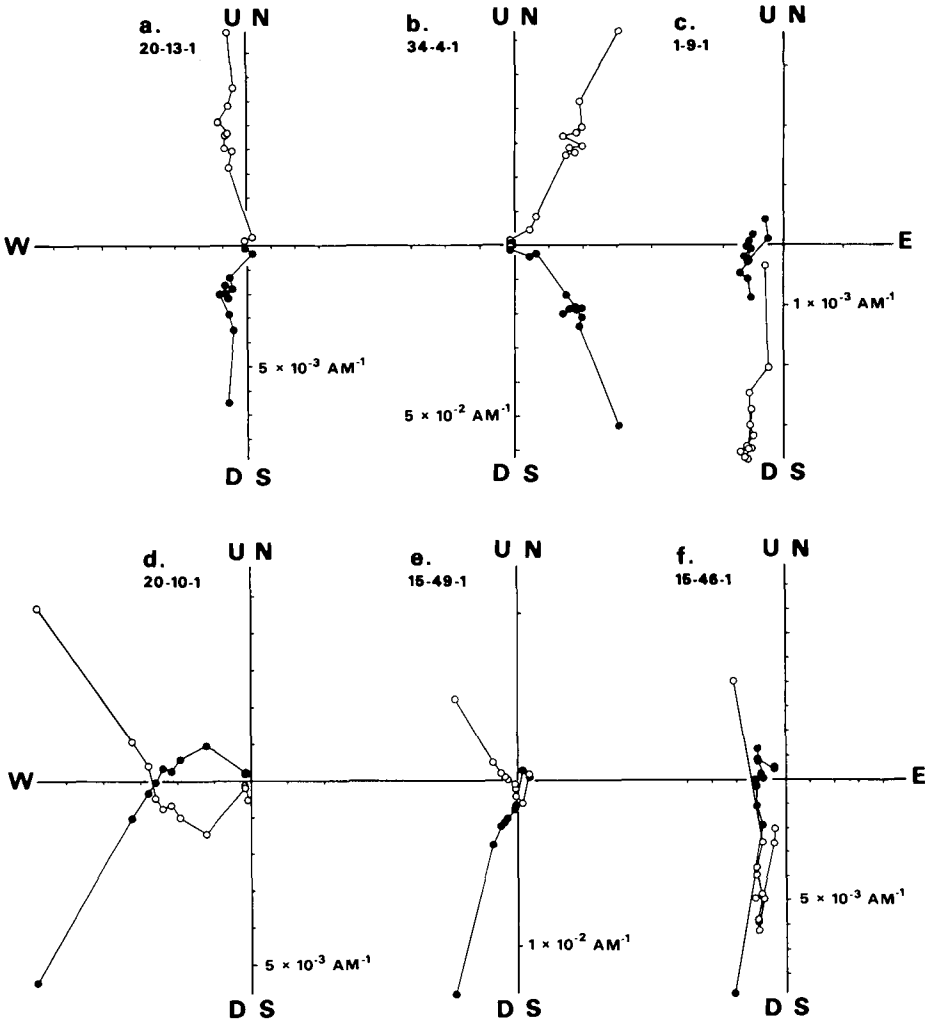


Figure 5. Representative Zijderveld plots of thermal demagnetization of red-bed samples, each in 11 or 12 steps—N, 450°, 550°, 570°, 590°, 610°, 630°, 650°, 665°, 675°, 685°, 700°. Sample numbers are shown on the plots. (a) and (b) show behavior of normal samples; (c) and (d), reversed; and (e) and (f), intermediate. Open and closed circles, respectively, denote horizontal and vertical components. All diagrams show remanent vector components without correction for bedding.

multiple normal and reversed components of Mesozoic remanence was shown by thermal demagnetization of any red specimen. Such evidence, in the form of flip-flopping directions occurring during chemical demagnetization, has been reported by Kluger and Sumner (1977) from coarse-grained samples of the Passaic Formation in the southwestern Newark Basin. (No chemical demagnetization was attempted in the present study because of the high clay content and associated low permeability of samples.)

The average direction of remanent vectors removed from red specimens by thermal demagnetization to 550 °C is nearly parallel to the recent normal axial dipole field direction of the study area (Fig. 4c). Apparently, this removed vector represents a normal secondary component acquired during Cenozoic time, possibly during the Brunhes normal epoch.

The remanence directions of red specimens demagnetized to 550 °C form distinct normal and reversed clusters (Fig. 4b). The average directions of these clusters (D = 7.0°, I = 25.9°, α_{95} = 3.2°; D = 190.9°, I = −1.0°, α_{95} = 5.4°; Table 2) are 153° apart rather than antipodal, probably due to incomplete removal of Cenozoic components by thermal demagnetization. The great circle joining the normal and reversed average directions passes within 2° of the recent axial dipole field direction (Fig. 4c).

Due to the generally uniform, gentle northwesterly dip of the Newark Basin strata, it was not possible to apply a fold test to the red-bed

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

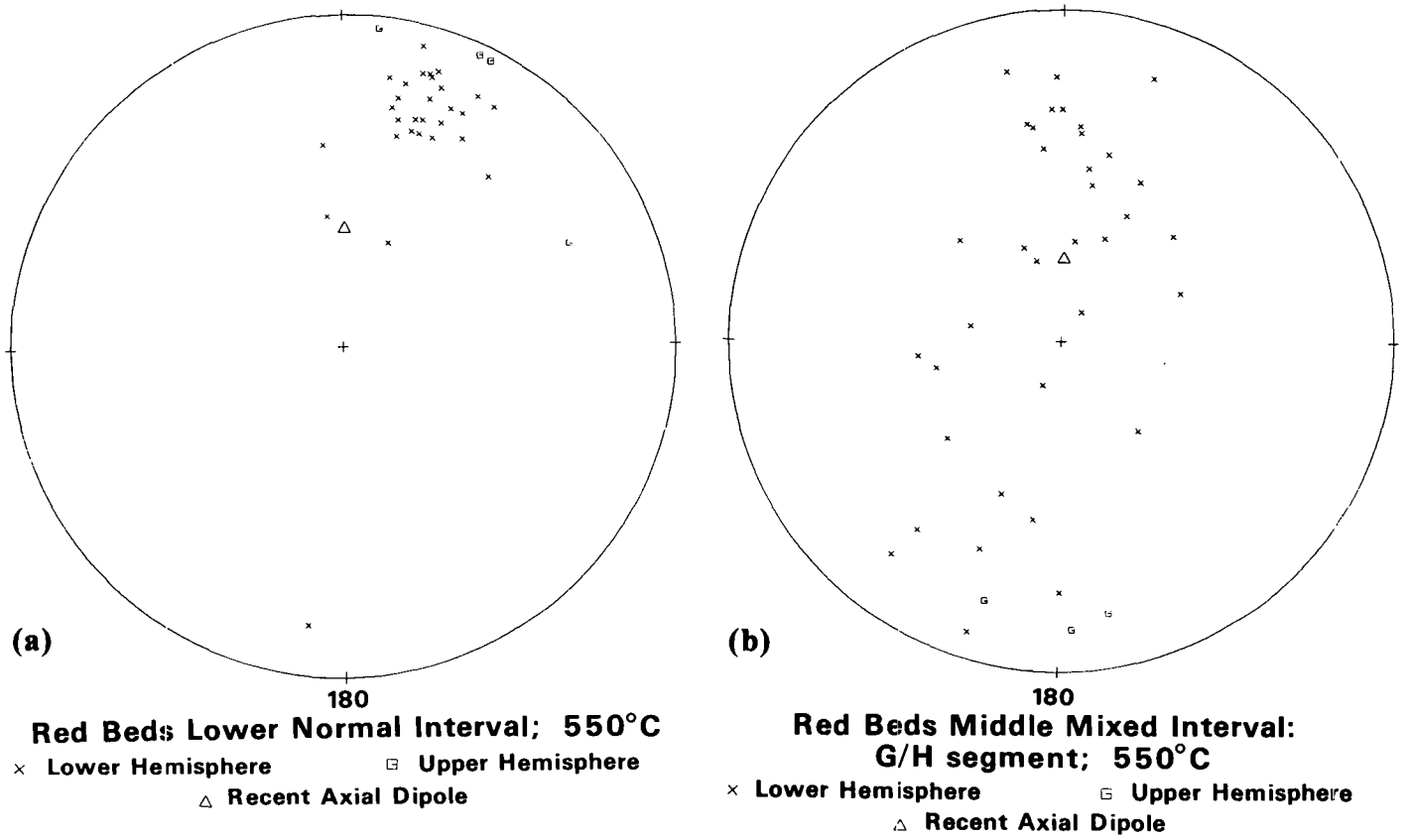


Figure 6. Equal-area projections of remanent vectors after 550 °C demagnetization for (a) lower normal interval, (b) middle mixed interval: G/H segment, (c) middle mixed interval: L/M/Perkasie segment, (d) middle mixed interval: Raritan/I-280 segment, (e) upper normal interval.

data. The application of structural corrections did not significantly affect the dispersion of normal and reversed remanence directions.

The average remanence directions of normal samples is in good agreement with paleomagnetic results reported from other Late Triassic/Jurassic rocks in North America (Table 4). The average remanence direction of reversed samples, however, is anomalously shallow.

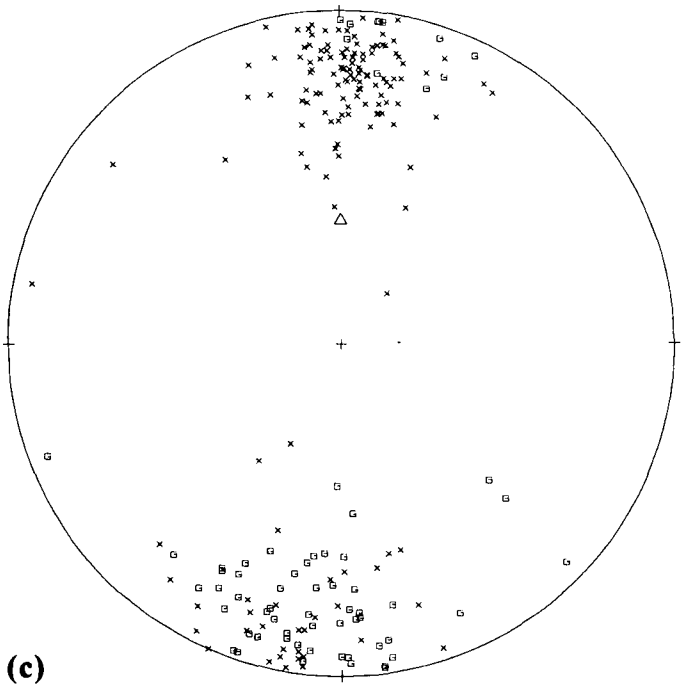
The polarity of each red-bed sample was classified by the following scheme. Data produced by the 550 °C demagnetization step were used for 427 samples and, for the remaining 24 samples, data from 450, 610, or 650 °C were used. Remanence directions that fell within 40° of the normal or reversed Late Triassic/Early Jurassic field directions (calculated from Newark igneous rocks) (Smith and Noltimier, 1979) were considered, respectively, normal (N) or reversed (R). Remanent vectors that fell between 40° and 90° of these axial dipole field directions were considered intermediate normal (IN) or intermediate reversed (IR). By these criteria, 59% of the 453 red outcrop samples are N, 28% are R, 8% are IN, and 5% are IR.

Alternating field (AF) demagnetization was performed on 12 red outcrop samples ranging in lithology from mudstone to sandstone. Peak fields of as much as 70 mT (700 oe) did not significantly affect the intensity or direction of remanence of the samples. A high coercivity is indicated, which is consistent with hematite as the carrier of remanence in the red beds.

Gray Outcrop Samples. Specimens of the 14 gray, lacustrine samples exhibited low NRM intensities (averaging 3×10^{-4} A/m) and their average NRM direction (D = 359.0°, I = 61.9°, α_{95} = 2.4°; uncorrected for bedding attitude) is parallel to the modern axial dipole field direction of the study area. AF demagnetization to peak fields of 70 mT reduced the average remanent intensity by half but did not change directions significantly. Thermal demagnetization to 450 °C produced low intensities and unstable directional behavior. It was concluded that the gray, lacustrine samples do not carry identifiable components of Mesozoic remanence; they were therefore rejected for paleomagnetic use. The gray members, however, still provide important stratigraphic control for interpretations of magnetostratigraphy, as discussed below.

Raritan Valley Subsurface Samples. Specimens from subsurface samples exhibit NRM directions and intensities that are generally within the range of values shown by specimens of red outcrop samples. The response to thermal demagnetization is also similar. Two features are particularly significant: (1) the average direction of the net vectors removed by thermal demagnetization to 550 °C is approximately parallel to the modern axial dipole field direction, although the degree of scatter is greater than that shown by vectors removed from outcrop specimens; and (2) increases in remanence intensity occurred during demagnetization of 90% of the reversed specimens but were not observed during demagnetization of normal specimens. Two inferences may be drawn from these features. First, red sedimentary rocks in the subsurface possess a Cenozoic normal remanence component similar to that of red beds in outcrop. This component, therefore, is not a result of surface weathering processes. Second, the orientation of the subsurface samples is less accurate than the orientation of outcrop samples but appears adequate for determination of magnetic polarity. For this reason, paleomagnetic data obtained from subsurface samples were not included in calculations of average remanence directions but were applied to interpretations of magnetostratigraphy, as de-

Downloaded from gsabulletin.gsapubs.org on October 27, 2010



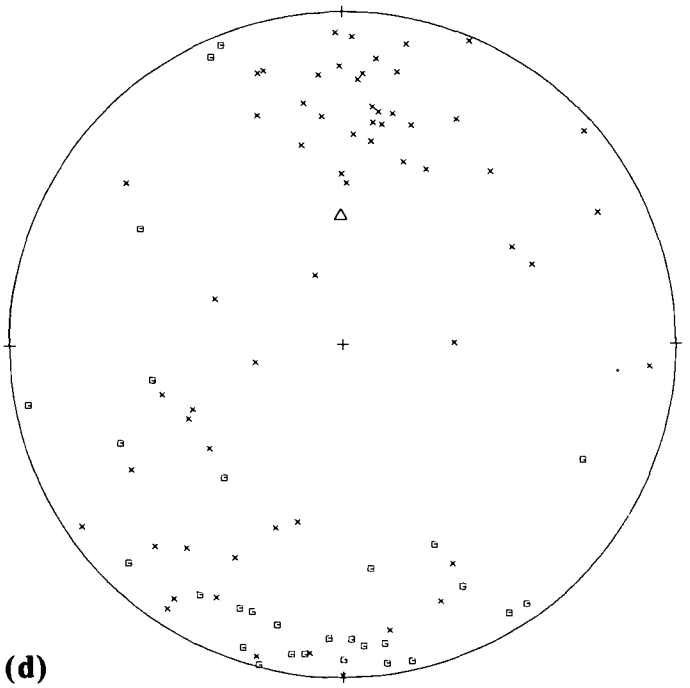


Figure 6 (*Continued*).

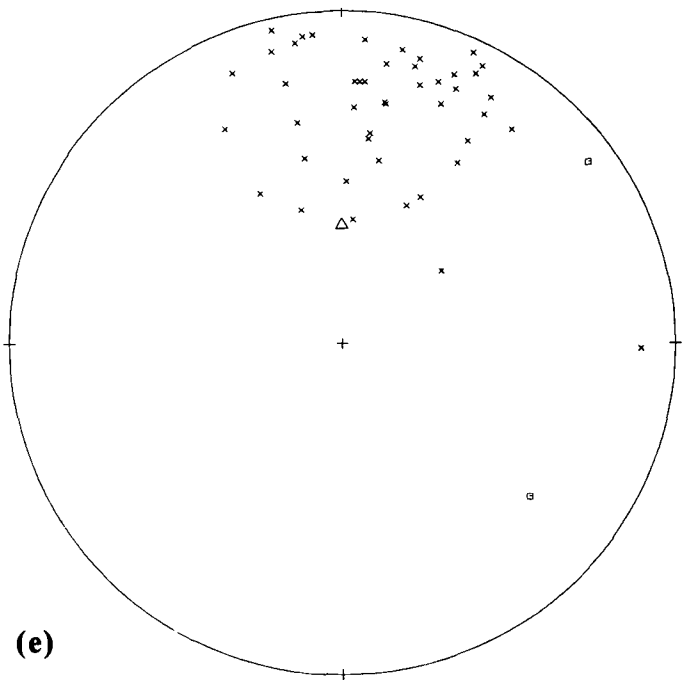
tailed below. According to the polarity criteria established above, 44% of the 153 red subsurface samples are N, 36% are R, 11% are IN, and 9% are IR.

Magnetostratigraphy

The magnetostratigraphy of the Newark Basin red beds can best be described by dividing the strata into three main stratigraphic intervals: a lower, 690-m interval of normal polarity, a middle, 2,600-m interval of mixed polarity, and an upper, 2,000-m interval of normal polarity (Fig. 2). The following sections present the data for each of these intervals.

Lower Normal Interval. Normal remanence directions were found in all samples of the upper Carnian sedimentary rocks within the uppermost Lockatong and lowermost Passaic Formations (Fig. 2; Table 1A). The remanence directions show a low degree of scatter (Fig. 6a; Table 2). Normal directions have also been previously reported from red members within and immediately above this stratigraphic interval (Du Bois and others, 1957; Opdyke, 1961).

Middle Mixed Interval. Alternating zones of normal and reversed polarity are present throughout most of the Passaic Formation (Fig. 2). The magnetostratigraphy of this mixed polarity interval is described below as three sep-



Downloaded from gsabulletin.gsapubs.org on October 27, 2010

arate segments. Evidence concerning the along-strike extent of polarity zones is also summarized.

1. G/H segment, central Newark Basin. A complex pattern of thin (0.5 to 20 m), normal and reversed zones is present in the 70-m-thick stratigraphic segment encompassing gray members G and H of the Passaic Formation (Fig. 3). The samples from three stratigraphically equivalent localities indicate that the polarity zones are discontinuous over a distance of 7.5 km along strike (Fig. 3, localities 4, 5, 6). The remanence directions of samples from this segment are poorly grouped (Fig. 6b; Table 2).

2. L/M/Perkasie segment, central Newark Basin. A simple pattern consisting of a normal polarity zone overlain by a reversed zone is present in the 550-m-thick stratigraphic segment represented by the uppermost exposures of the Passaic Formation (Fig. 3) above the G/H segment. The normal polarity zone encompasses gray members L and M and extends along strike for at least 15 km (Fig. 3, localities 8, 9, 10, 11). The normal to reversed polarity boundary can be traced along strike for at least 52 km and is situated at a stratigraphic depth of 10 to 25 m below the base of the Perkasie member (Fig. 3, localities 12, 13, 14, 15). The depth increases regularly toward the northeast, and this trend parallels the northeastward increase in thickness of the red member underlying the Perkasie member (Fig. 3; McLaughlin, 1959). At some localities, thin (0.5 to 2 m) anomalous zones of intermediate or normal polarity are intercalated in the reversed strata immediately above the polarity transition or immediately below the base of the Perkasie member. These anomalous zones are not laterally continuous along strike. The remanence directions of the samples from the L/M/Perkasie segment are generally tightly grouped (Fig. 6c; Table 2).

3. Raritan/I-280 segment, northeastern Newark Basin. Complex magnetostratigraphic patterns are indicated by the outcrop and subsurface samples from all but the uppermost 165 m of the Passaic Formation (Fig. 7). Where sampling was significantly dense and continuous, the data suggest the presence of thick (>75 m) polarity zones interrupted by thin (0.5 to 20 m) anomalous zones of opposite or intermediate polarity (Fig. 8). Most of these anomalous zones are stratigraphically adjacent to gray lacustrine members. A comparison between outcrop data and data from subsurface samples 2 km to the north (Fig. 8) suggests that the thick zones are continuous but the thin zones are discontinuous along strike. Both sections exhibit predominantly normal polarity at the base and predominantly reversed polarity at the top; the polarity transition in the subsurface is situated within 15 m of the stratigraphic position of the

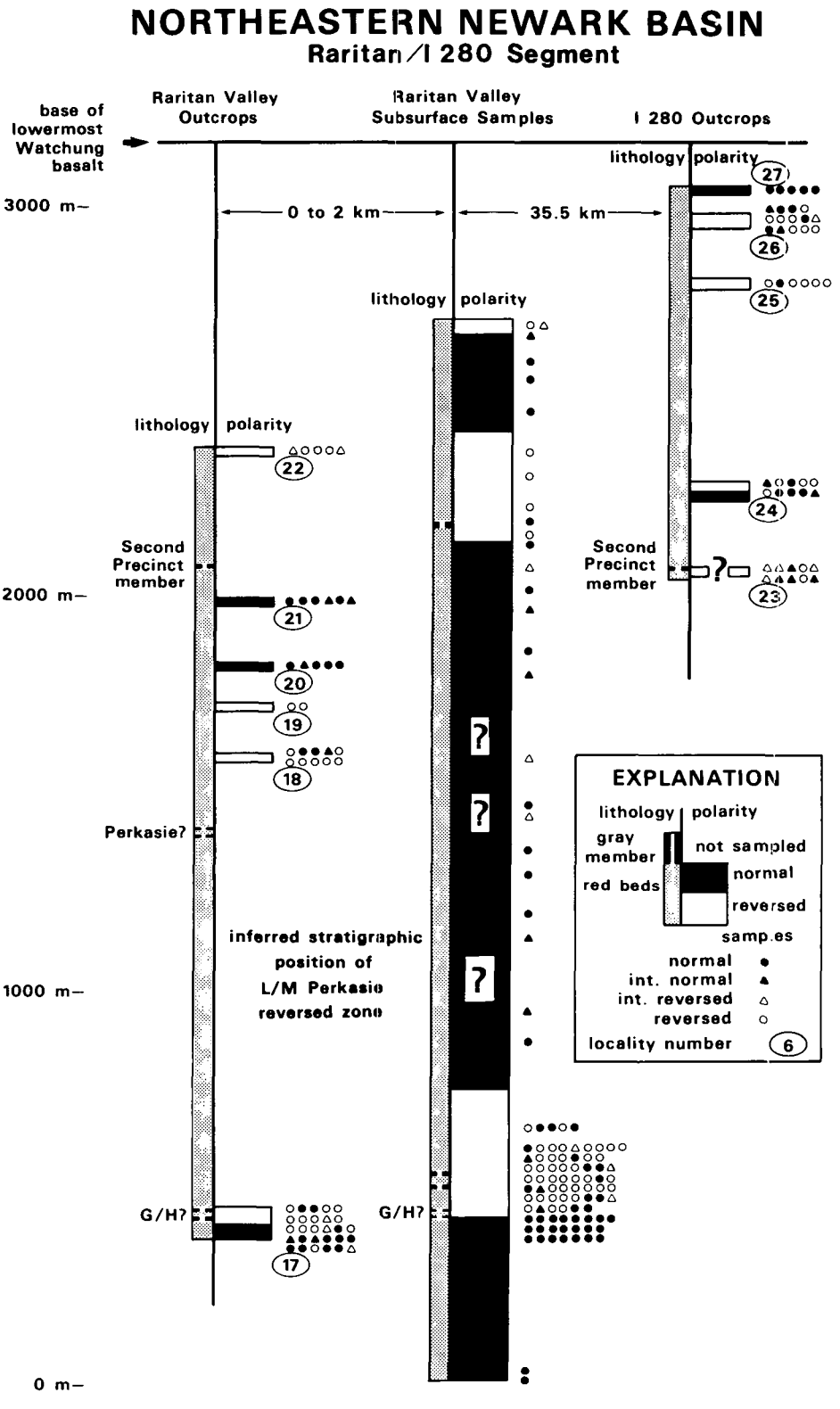


Figure 7. Data from the Raritan/I-280 segment of the Passaic Formation. Note generally complicated magnetostratigraphy, abundance of intermediate directions, and lack of clear along-strike magnetostratigraphic correlation of individual polarity zones. Data detailed in Figure 8 have been generalized here. The inferred stratigraphic position of the reversed zone of the L/M/Perkasie segment, as shown, is based on paleontologic correlations (P. E. Olsen, 1980, written commun.).

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

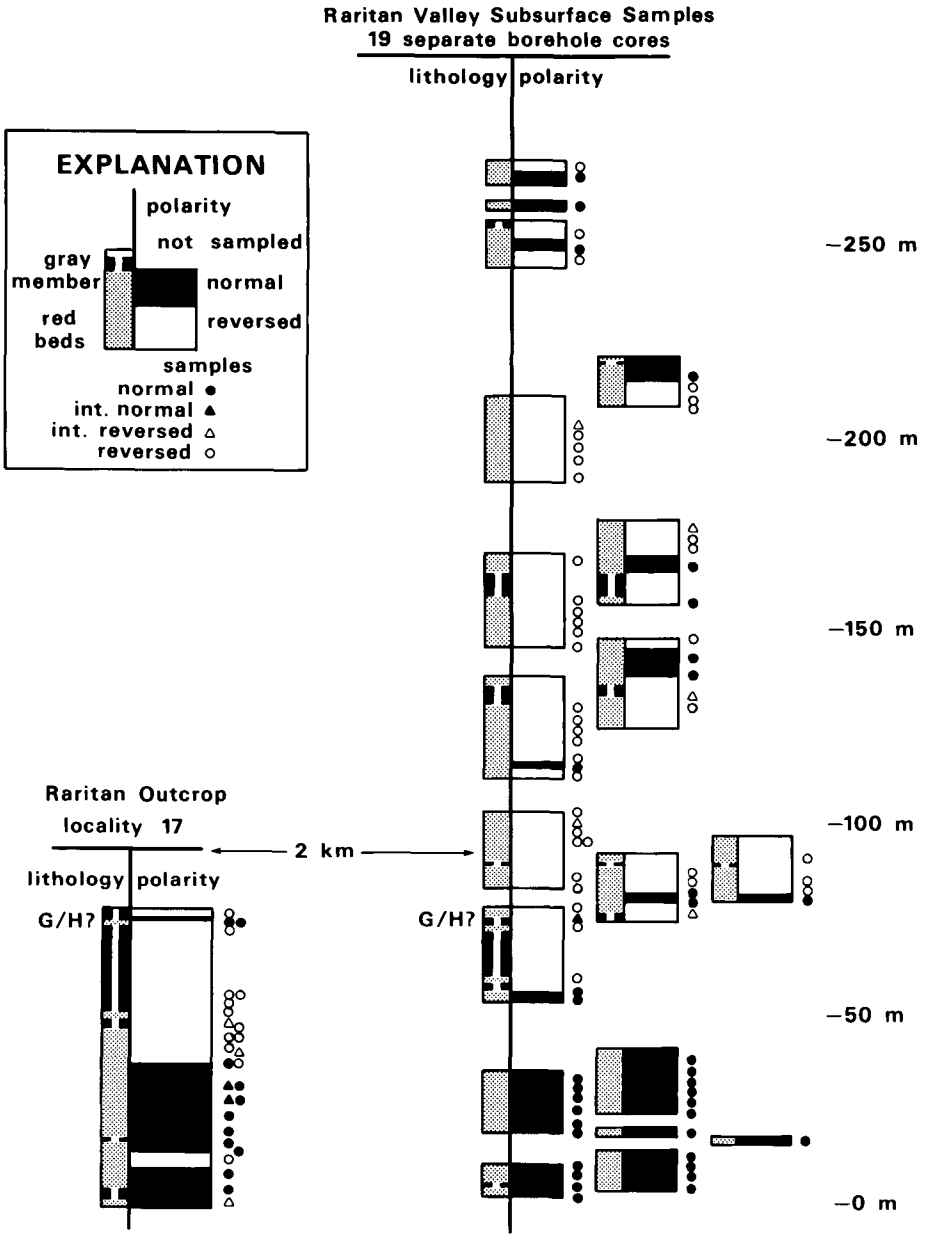


Figure 8. Detail of the lower part of Figure 7, comparing data from outcrop locality 17 and correlative subsurface samples. The general normal-over-reversed polarity zonation is complicated by the thin anomalous zones which are associated with gray members.

transition in outcrop (Fig. 8). Some of the thin anomalous zones shown by subsurface samples are traceable between overlapping sections of adjacent boreholes (separated laterally by 100 to 200 m), but none can confidently be correlated with the outcrop section (Fig. 8). The outcrop samples from the Raritan/I-280 segment show generally poor grouping of remanence directions (Fig. 6d; Table 2).

The stratigraphic correlation between the central and northeastern Newark Basin sections is incompletely understood. On the basis of paleontologic criteria (P. Olsen, 1980, written commun.), a tentative correlation can be made between gray members G and H of the central Newark Basin and a 20-m-thick gray member identified in outcrop and subsurface in the northeastern Newark Basin (Figs. 2 and 7). By this correlation, the G/H and L/M/Perkasie segments are respectively equivalent to the base and middle of the Raritan/I-280 segment. The magnetostratigraphic patterns are not clearly consistent with this paleontologic correlation (Fig. 2). The base of the Raritan/I-280 segment shows predominantly reversed polarity, whereas the G/H segment shows mixed polarity. Furthermore, the thick (>200 m), reversed zone in the L/M/Perkasie segment was not identified in the Raritan/I-280 segment. These discrepancies may reflect incorrect paleontologic correlation or, alternatively, incomplete sampling. The part of the Raritan/I-280 segment supposedly correlative with the L/M/Perkasie reversed zone is represented by two samples only, both showing intermediate-normal polarity (Fig. 7).

Upper Normal Interval. Normal to intermediate-normal directions were shown by all samples of the upper Rhaetian to Sinemurian red beds within the uppermost Passaic Formation and the Feltville, Towaco, and Boonton Formations (Fig. 2; Table 1B). The remanence directions were moderately well grouped (Fig. 6e; Table 2).

PALEOMAGNETISM OF THE BASALTS

One or more specimens from each of the 120 samples of the Watchung basalts were subjected to step-wise, progressive AF demagnetization in peak fields of as much as 30 mT (300 oe). Demagnetization lowered remanent intensities by 50% to 90% and significantly reduced the scatter of remanent vectors (Fig. 9). In most cases, within-site scatter was minimized by demagnetization to 20 mT. The average direction of vectors removed by demagnetization fell between the Early Jurassic and modern axial dipole field directions, probably reflecting removal of recently acquired viscous remanent magnetization (VRM) components.

The main paleomagnetic features of the Watchung basalts (as detailed in Table 3) can be summarized as follows. (1) All samples showed normal polarity, in agreement with the polarity of intercalated red beds in the "upper normal interval." (2) Within-site scatter was low (α_{95} averages 7.5°). (3) Following demagnetization to 20 mT, the mean remanence direction of the basalt samples (D = 7.2, I = 27.2, α_{95} = 2.3) was in close agreement with the average direction of normal Newark Basin red-bed samples (Table 2), and its average VGP is consistent with published Early Jurassic paleopole data (Table 4). (4) The mean remanence directions of the three Watchung basalts did not differ significantly. The two to three lava flows that comprise each formation are, however, considered too few to fully average the effects of paleosecular variation. (5) Samples from the base of the Preakness (Second Watchung) Basalt (Table 3, sites, 6, 7, 8) revealed a stratigraphically systematic variation in remanence directions which may represent a record of paleosecular variation

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

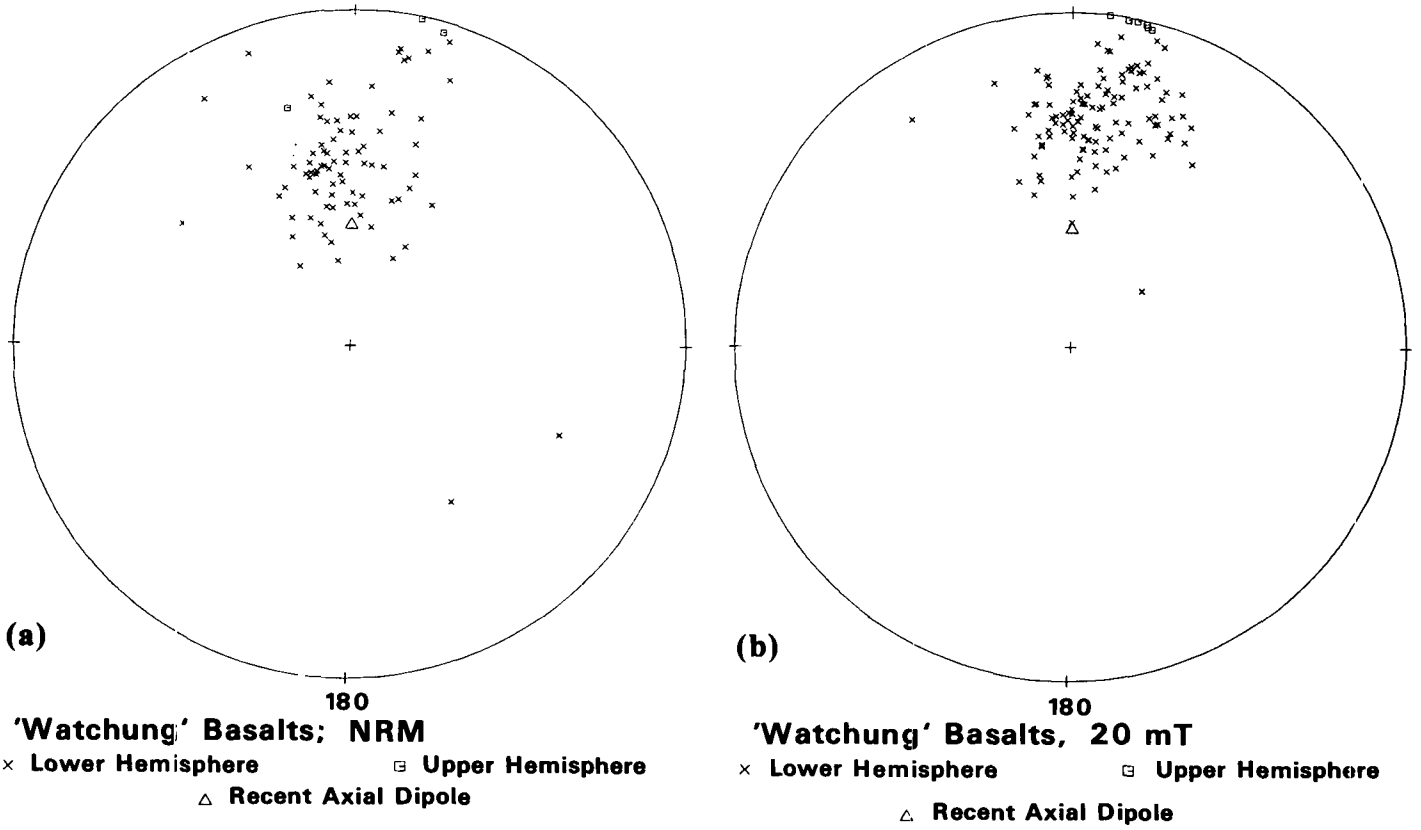


Figure 9. Equal-area projections of remanent vector data from the basalt samples (a) = NRM and (b) = 20 mT. NRM was not measured in all samples.

during cooling of the lava flow (McIntosh, unpub. data). The first four of these features are in accord with the conclusions of a previous paleomagnetic study of the Watchung basalts (Opdyke, 1961).

CORRELATION

Available paleomagnetic polarity data from well-dated Late Triassic to Early Jurassic rocks in North America, Africa, and the Soviet Union are summarized in Figure 10. Detailed correlation of data from sedimentary basins in eastern North America and northern Africa is facilitated by recent biostratigraphic and radiometric age determinations (Olsen and others, 1982; Manspeizer and others, 1978).

Early Jurassic

Paleomagnetic studies of Lower Jurassic rocks in eastern North America have shown normal polarity only, in good agreement with the upper normal interval of the Newark Basin (Fig. 10). Units showing normal polarity include Hettangian to Sinemurian basaltic lavas and intrusive rocks (deBoer, 1968; Smith and Noltimier, 1979; Irving and Banks, 1961; Bowker, 1960) and upper Rhaetian to Pleinsbachian, CRM-bearing red beds (Brown, 1979; and 1982, personal commun.; Kelly, 1979; Amerigian and Watkins, 1977). The VGP's reported from these units are also in generally good agreement with Newark Basin data (Table 4); detailed correlations based on the remanence directions of individual lava units (deBoer, 1968; Manspeizer and others, 1978; Olsen and others, 1982) are not considered reliable because of the small number of flows in each unit (typically two to four), insufficient to average out secular variation.

Paleomagnetic data from North Africa also correlate well with the upper normal interval of the Newark Basin (Fig. 10). Moroccan rocks of normal polarity include numerous upper Rhaetian to Sinemurian basaltic lavas (Martin and others, 1978; Daly and Pozzi, 1976; Hailwood, 1975; Bardon and others, 1973), Lower Jurassic dikes (Hailwood and Mitchell, 1971), and a Lower Jurassic limestone (Martin and others, 1978). In addition, a composite polarity scale compiled from U.S.S.R. data shows normal polarity during Early Jurassic time (Perchevsky and Kramov, 1973).

Whether the geomagnetic field was normal throughout all of Early Jurassic time is uncertain. Results from basaltic intrusive rocks in eastern North America indicate a period of predominantly normal polarity in latest Early Jurassic time (Smith and Noltimier, 1979). The results from Karoo basalts (van Zijl and others, 1962) and Nuanetsi intrusive rocks (Gough and others, 1964), however, suggest a period of mixed polarity in middle Early Jurassic time. The Pleinsbachian(?) Kayenta Formation of the southwestern United States also shows mixed paleomagnetic polarity (Steiner and Helsley, 1974; age according to Olsen and others, 1982). In addition, reversed polarity has been reported from a few isolated igneous units of possible Lower Jurassic age, including dikes in Massachusetts (Smith, 1978) and Rhode Island (Choinere and Watkins, 1976) and one basalt flow in Morocco (Martin and others, 1978).

Late Triassic

Mixed polarity has been shown by most studies of Upper Triassic sedimentary rocks in North America (Fig. 10). These data correlate well with the Newark Basin's Norian to late Rhaetian mixed polarity interval but correlate poorly with the underlying late Carnian normal interval. Smith (1978) made a reconnaissance study of

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

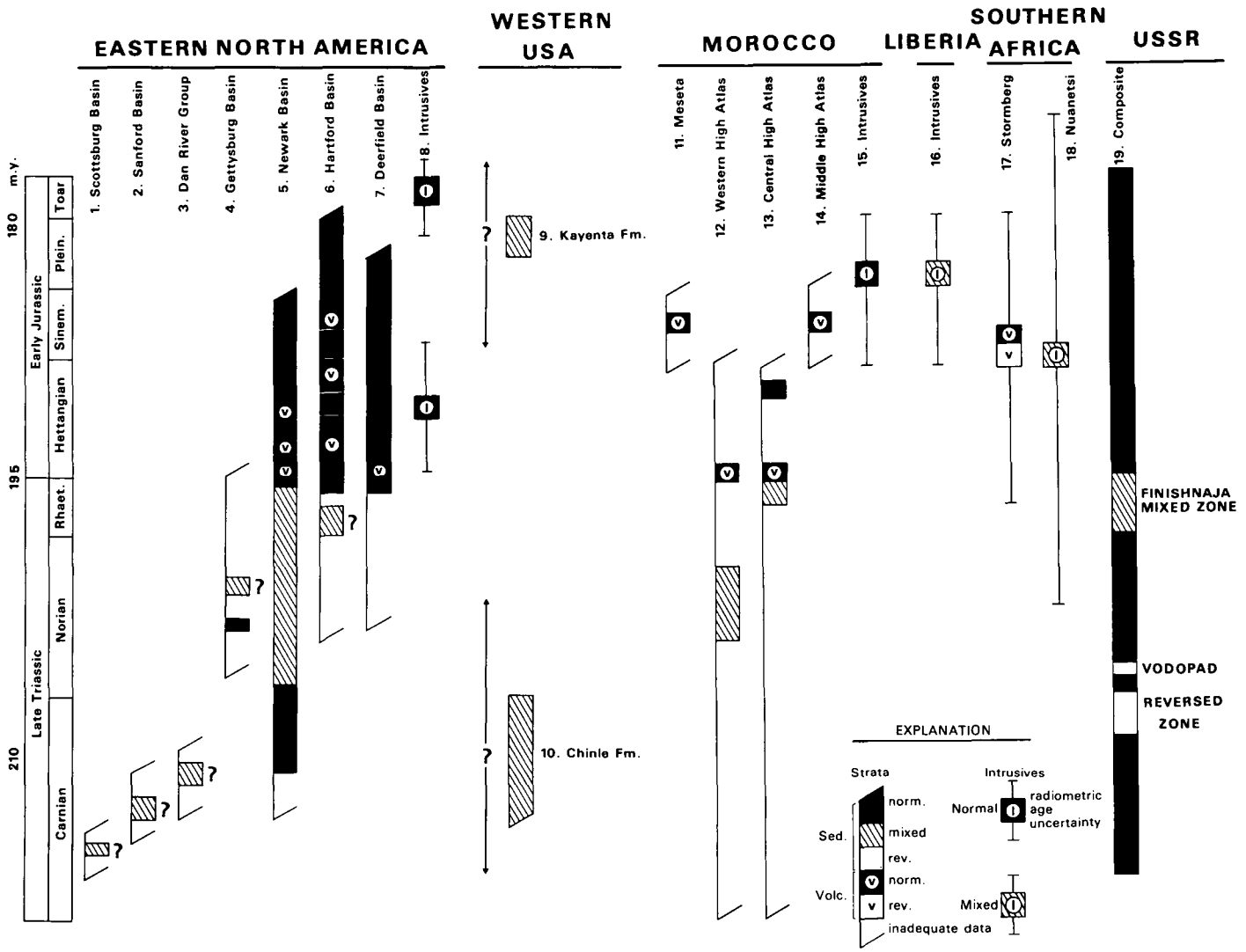


Figure 10. Correlation of paleomagnetic data from North America, Africa, and the U.S.S.R. The biostratigraphic ages of North American and Moroccan units are after Olsen and others (1982) and Manspeizer and others (1978). Sources of paleomagnetic data: (1), (2), and (3) Bowker (1960); (4) Smith (1978; and 1982, personal commun.); (5) this study; (6) and (7) Brown (1979; and 1982, personal commun.), Smith and Noltimier (1979), Smith (1978; and 1982, personal commun.); (8) Smith and Noltimier (1979); (9) Steiner and Helsley (1974); (10) Reeve and Helsley (1972); (11) Daly and Pozzi (1976, locality M), Bardon and others (1973, localities 6 and 7); (12) Bardon and others (1973, locality 5), Martin and others (1978, locality M12); (13) Bardon and others (1973, localities 1, 2, 3); Hailwood (1975); Martin and others (1978, localities M11 and M14); (14) Daly and Pozzi (1976, locality C); (15) Hailwood and Mitchell (1971); (16) Dalrymple and others (1975); (17) van Zijl and others (1962), radiometric age after Fitch and Miller (1971); (18) Gough and others (1964); (19) Perchevsky and Kramov (1973) (note: rock types were not detailed in U.S.S.R. data).

the Norian to Rhaetian New Haven, New Oxford, and Gettysburg Formations of Connecticut, Delaware, and Maryland. Thermal and AF demagnetization revealed mostly scattered or anomalous results, some of which he considered "may be a poorly preserved record of superimposed normal and reversed Norian-Rhaetian magnetization" (Smith, 1978, p. 272). Bowker (1960) reported anomalous NRM directions from a small number of Carnian red-bed samples from the Scottsburg, Sanford, and Dan River Basins of Virginia and North Carolina. In addition, the late Carnian(?) Chinle Formation of New Mexico shows mixed polarity (Reeve and Helsley, 1972; age according to Olsen and others, 1982). Other Upper Triassic formations from the southwestern United States, including the Navajo and Wingate Sandstones, have been rejected for paleomagnetic use (Johnson, 1976; Reeve and Helsley, 1974). Volcanic units of Carnian to mid-Rhaetian age are not known from North America.

Paleomagnetic results from Upper Triassic rocks in Morocco and the U.S.S.R. are in good agreement with the Norian to late Rhaetian mixed polarity interval of the Newark Basin (Fig. 10). Mixed polarity has been reported from two studies of Norian to Rhaetian red beds in Morocco (Martin and others, 1978; Hailwood, 1975). Upper Carnian sedimentary rocks are present in Morocco (Manspeizer and others, 1978) but have not been investigated paleomagnetically. The composite polarity scale from the

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

U.S.S.R. also shows a period of mixed polarity in Rhaetian time. Two reversed intervals are, however, indicated in late Carnian time (Fig. 10; Perchevsky and Kramov, 1973), correlating poorly with the Newark Basin's late Carnian normal interval.

DISCUSSION

Acquisition of Remanence in Newark Basin Red Beds

Petrographic data suggest that the remanence of the Newark Basin red beds is a chemical remanent magnetization (CRM). The red-bed samples contain a variety of forms of hematite and are devoid of magnetite and titanomagnetite. Most of the hematite is in the form of authigenic, translucent microcrystallites, the majority of which are disseminated in the clays. Less abundant specular hematite makes up ~1% of the volume of each sample. More than half of the specular hematite is clearly authigenic and <5% is diagnostically detrital. All of the observed forms of hematite are larger than the superparamagnetic threshold (0.03 μ; Banerjee, 1971) and hence carry remanence. The relative volumes of authigenic and detrital hematite therefore suggest that the major portion of remanence is a CRM and that components of detrital remanence (DRM) are minor.

The timing of the development of the different forms of authigenic hematite probably varied widely. Red pigment has been observed in sediments as young as tens of thousands of years (Walker, 1967). At the other extreme, alteration of ferromagnesian silicates to hematite may have taken tens of millions of years (Larson and Walker, 1975). Acquisition of CRM in the Newark Basin red beds probably began soon after deposition and continued for millions of years. The relative magnitudes of early- and late-formed CRM components, however, cannot be determined from the petrographic data.

Paleomagnetic data are consistent with the interpretation that the remanence of the Newark Basin red beds is primarily a CRM. Furthermore, the data shed additional light on the timing of remanence acquisition. The red beds apparently carry both Late Triassic/Early Jurassic and Cenozoic components of remanence.

Late Triassic/Early Jurassic remanence components are indicated by the vectors remaining after partial demagnetization. The average VGP of demagnetized normal red-bed samples is nearly coincident with the average VGP of the intercalated Watchung basalts and is, moreover, in agreement with published North American Late Triassic/Early Jurassic paleopole data (Table 4). The red beds apparently acquired these remanence components prior to the Late Jurassic, by which time the geomagnetic pole had shifted considerably eastward (Morel and Irving, 1981).

The magnetostratigraphic polarity pattern of thermally demagnetized red-bed samples suggests that their remanence primarily represents CRM components acquired within a few million years or less after deposition. The gross pattern shown by the red beds is late Carnian normal, Norian to late Rhaetian mixed polarity, and late Rhaetian to Sinemurian normal. The Lower Jurassic normal red beds were certainly deposited during a period of normal geomagnetic polarity, as shown by paleomagnetic results from correlative igneous rocks in eastern North America and Morocco (Fig. 10; Smith and Noltimier, 1979). This period of normal polarity was followed by reversed or mixed polarity in Pleinsbachian time, as indicated by African paleomagnetic data (Fig. 10). Apparently, the Lower Jurassic red beds in the Newark Basin acquired a CRM *sufficient to establish polarity* within, at most, a few million years prior to the Pleinsbachian geomagnetic reversals. If, instead, polarity had been established tens of millions of years after deposition, it is unlikely that the 2,000-m-thick sequence of impermeable mudstone would have acquired uniformly normal polarity. In contrast, similar thick sequences of normal polarity in Jurassic and Miocene sandstones of the western United States have been interpreted as evidence of recent magnetization (Johnson, 1976; Hoblitt and others, 1974), but the remanence directions of the Newark Basin red-bed samples preclude such recent remagnetization.

Data from the Norian to upper Rhaetian red beds in the mixed polarity interval suggest that CRM acquisition was not penecontemporaneous with deposition but was relatively rapid. The mixed polarity interval shows a pattern of thick (as much as 200 m), alternating normal and reversed zones, in places exhibiting numerous thin (0.5 to 20 m), anomalous zones of opposite or intermediate polarity. The thinness, limited along-strike extent, and frequency of these anomalous zones indicate that they are not accurate records of the geomagnetic field at the time of their deposition. Furthermore, the association of these anomalous zones with gray, lacustrine units implies lithologic control of the timing of remanence acquisition.

Evidence for relatively rapid CRM acquisition is provided by the boundaries between the thick polarity zones in the mixed polarity interval. These boundaries are not associated with lithologic changes and one boundary can be traced along strike for at least 52 km. Variations in the stratigraphic position of this polarity boundary are slight and may reflect local variations in sedimentation rate. Apparently, the acquisition of CRM at this boundary was closely synchronized over a wide area. Given the locally variable nature of diagenetic processes (Walker and others, 1981), such synchronization supports relatively rapid acquisition of CRM components; polarity may have been established within hundreds of thousands of years. By comparison, regionally extensive polarity-zone boundaries in the Moenkopi Formation have been interpreted as being products of basin-wide diagenetic events controlled by ancient groundwater tables (Larson and others, 1982); it is unlikely that such events occurred in the low-permeability mudstones of the Newark Basin.

In addition to Late Triassic/Early Jurassic CRM components, the Newark Basin red beds apparently carry removable and unremovable, late-formed, Cenozoic or Mesozoic remanence components. Removable Cenozoic components are indicated by the steep normal vectors which are, on the average, parallel to the recent axial field. These were lost from red-bed samples during thermal demagnetization. The presence of removable Cenozoic components in both outcrop and subsurface samples suggests that they are not related to surficial weathering processes. These components probably represent a viscous remanent magnetization (VRM) acquired during the Brunhes normal epoch, as suggested by their "hardness" to AF demagnetization and their "softness" to thermal demagnetization (Dunlop and Stirling, 1977). Unremovable Cenozoic remanence components are suggested by the nonantipodal average remanence directions of normal and reversed demagnetized samples, the latter being anomalously shallow (Fig. 4b). This feature could alternatively be attributed to inclination error related to DRM on compaction, but this explanation is rejected. Such inclination errors should flatten normal and reversed Triassic/Jurassic vectors equally, whereas addition of steep normal components would flatten reversed vectors and, to a lesser extent, steepen normal vectors. The mechanism by which the unremovable Cenozoic components were acquired is uncertain; components of VRM carried by hematite should have been

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

completely removed by thermal demagnetization to 375 °C (Dunlop and Stirling, 1977).

Two additional features suggest the presence of unremovable, late-formed remanence components in the red-bed data. First, the VGP's of the red-bed samples vary stratigraphically but do not show any progressive trend from the base to the top of the section (Table 2). If the red-bed data were actually free of late-formed components, it would have provided a systematic record of movement of the North America plate in Late Triassic and Early Jurassic time. Second, 14% of the samples exhibit intermediate directions, and most of these show low intensities. Many of these intermediate samples were situated near polarity-zone boundaries or adjacent to gray, lacustrine units (Figs. 3, 7, 8). The incidence of these samples may reflect the presence of multiple, anti-parallel, Late Triassic/Early Jurassic remanence components. Such components were not detected in the course of thermal demagnetization but may reside in minerals with similar blocking-temperature spectra (Larson and others, 1982).

On the basis of paleomagnetic and petrographic data, we propose the following model of remanence acquisition in the Newark Basin red beds:

1. Most or all of the remanence was acquired as a CRM during diagenesis.

2. Components of CRM which were sufficient to establish polarity were acquired within hundreds of thousands to a few million years after deposition. Deposition and magnetization within long periods (>5 m.y.) of predominantly normal polarity produced thick sequences of predominantly normal red beds—the late Carnian and late Rhaetian to Sinemurian normal intervals. Deposition and magnetization within a period of alternating polarity produced a red-bed sequence of mixed polarity—the Norian to late Rhaetian mixed interval. Because of local variations in lithology and rates of diagenesis, the polarity-zone sequence within the mixed polarity interval is not expected to mirror accurately the geomagnetic field behavior at the time of deposition.

3. Further diagenetic processes during Mesozoic and Cenozoic time added additional components of remanence. Although the Cenozoic components can be largely removed by thermal demagnetization, some unremovable, late-formed components remain.

If the above model is correct, then the gross pattern of magnetostratigraphy shown by the Newark Basin red beds—late Carnian to Norian normal, Norian to late Rhaetian mixed polarity, late Rhaetian to Sinemurian normal—reflects the general behavior of the Late Triassic/Early Jurassic geomagnetic field. The sequence of individual polarity zones may not, however, correspond to the actual sequence of geomagnetic polarity reversals at or near the time of deposition.

The above model of remanence acquisition differs strongly from that proposed by Purucker and others (1980) for the Moenkopi Formation of the Western United States, in that we conclude that primary remanence components cannot confidently be isolated from the NRM of the red beds. The above model is similar to that proposed by Larson and others (1982) for the Moenkopi Formation, except that we believe that the major part of the remanence was acquired within hundreds of thousands to a few million years rather than tens of millions of years. Hematite authigenesis and consequent remanence acquisition may have proceeded more rapidly in the clay-rich, fine-grained Newark Basin units than in the coarser Moenkopi sandstones.

Application of Newark Basin Paleomagnetic Data

In spite of the delayed nature of remanence acquisition in the Newark Basin red beds, their gross magnetostratigraphic polarity pattern may be useful for correlation with rocks elsewhere. The most distinctive datum is the upper boundary of the mixed polarity interval. This change from mixed to normal polarity was found in red beds of late Rhaetian age (~197 m.y.) and is supported by paleomagnetic data from outside the Newark Basin (Fig. 10). This datum could prove to be a valuable adjunct to radiometric and biostratigraphic age dating of Upper Triassic rocks. Because the mixed polarity interval and overlying normal interval represent time periods of more than ten million years, the mixed to normal polarity change should be recorded even by sedimentary rocks which acquired remanence over a few millions of years.

A second potentially useful feature for correlation is the lower boundary of the Newark Basin's mixed polarity interval. Data from the red beds suggest a late Carnian period of normal polarity changing to mixed polarity at the end of Carnian time. Available data from sedimentary units outside the Newark Basin do not, however, support a long late Carnian period of normal polarity. This discrepancy may reflect insufficient data, inaccurate biostratigraphic ages, or a period of normal polarity which was shorter than the duration of remanence acquisition in some of the units.

It is unlikely that individual polarity zones within the Newark Basin's mixed polarity interval will prove to be useful for correlation with other Upper Triassic rocks. As previously discussed, some polarity-zone boundaries are continuous along strike, reflecting remanence acquisition that was rapid relative to the timing of geomagnetic reversals. Other polarity zones are discontinuous, indicating locally variable remanence acquisition spanning two or more polarity intervals. It would be difficult or impossible to decipher the actual sequence of Late Triassic geomagnetic field reversals from the data from the mixed polarity interval.

Caution is indicated in the application of Newark Basin red-bed data to tectonic reconstructions. Accurate reconstructions require data that are free from unremovable, late-formed remanence components. Such components are certainly present in the Newark Basin data, as shown by the variable VGP's which lack a systematic stratigraphic trend but which are systematically different between normal and reversed samples (Table 2). Apparently, the unremovable, late-formed components are large enough to obscure any progressive stratigraphic paleomagnetic record of movement of the North American plate.

ACKNOWLEDGMENTS

Many thanks to Naoma Dorety, who helped in most phases of this project, including measurement, data reduction, and encouragement. Anne Demitrack was responsible for data from red bed locality 11. Thanks to the many friends who generously assisted in the field. Paul Olsen provided valuable information on stratigraphy and location of outcrops.

We appreciate critical reviews of this paper by F. B. Van Houten, M. E. Purucker, N. D. Opdyke, Lisa Tauxe, N. Triffleman, J. R. Keys, W. Manspeizer, and E. E. Larson.

The Raritan River borehole cores were drilled by the Middlesex County Sewerage Authority in 1975. Permission to study and sample these cores was provided by the Executive Director, Anthony J. Popowski. This work was supported in part by National Science Foundation Grants GA-41699 and EAR 8008207.

Downloaded from gsabulletin.gsapubs.org on October 27, 2010

REFERENCES CITED

Ade-Hall, J. M., Khan, M. A., Dagley, P., and Wilson, R. L., 1968, A detailed opaque petrological and magnetic investigation of a single Tertiary lava flow from Skye, Scotland; I, Iron-titanium oxide petrology: Royal Astronomical Society Geophysical Journal, v. 16, p. 374–388.

Amerigian, C., and Watkins, N. D., 1977, Paleoenvironmental control of primary and secondary components of magnetization in the East Berlin Formation of Connecticut [abs.]: EOS (American Geophysical Union Transactions), v. 58, p. 728.

Baag, C., and Helsley, C. E., 1974, Evidence for penecontemporaneous magnetization of the Moenkopi Formation: Journal of Geophysical Research, v. 79, p. 3308–3319.

Banerjee, S. K., 1971, New grain size limits for paleomagnetic stability in hematite: Nature; Physical Science, v. 232, p. 15–16.

Bardon, C., Brossert, A., Hamzeh, R., and Westphal, M., 1973, Etude paleomagnetique de formations du Trias et du Jurassique du Maroc et du Sahara: Compte Rendue Academie des Science, Paris, v. 276, p. 2356–2360.

Blatt, H., and Southerland, B., 1969, Interstratal solution and nonopaque heavy minerals in shales: Journal of Sedimentary Petrology, v. 39, p. 591–600.

Bowker, D. E., 1960, Remanent magnetization of eastern United States Triassic rocks [Ph.D. thesis]: Cambridge, Massachusetts, Massachusetts Institute of Technology.

Brown, L., 1979, Magnetic observations of the red beds of the northern Connecticut Valley—DRM or CRM? [abs.]: EOS (American Geophysical Union Transactions), v. 60, p. 815.

Choinere, S. R., and Watkins, N. D., 1976, Paleomagnetism of some Rhode Island area dikes: Geological Society of America Abstracts with Programs, v. 8, p. 151.

Cornet, B., 1977, The palynostratigraphy and age of the Newark Supergroup [Ph.D. thesis]: University Park, Pennsylvania, Pennsylvania State University, 506 p.

Cornet, B., and Traverse, A., 1975, Palynological contributions to the chronology and stratigraphy of the Hartford Basin in Connecticut and Massachusetts: Geoscience and Man, v. 11, p. 1–33.

Dalrymple, G. B., Gromme, S. C., and White, R. W., 1975, Potassium-argon age and paleomagnetism of diabase dikes in Liberia: Initiation of Central Atlantic rifting: Geological Society of American Bulletin, v. 86, p. 399–411.

Daly, L., and Pozzi, J. P., 1976, Resultats paleomagnetiques du Permian Inferior et du Trias Marocain: Comparison avec les donnees Africaines et Sud Americaines: Earth and Planetary Science Letters, v. 29, p. 71–81.

deBoer, J., 1968, Paleomagnetic differentiation and correlation of the Late Triassic volcanic rocks in the central Appalachians (with special reference to the Connecticut Valley): Geological Society of America Bulletin, v. 79, p. 609–625.

Du Bois, P. M., Irving, E., Opdyke, N. D., Runcorn, S. K., and Banks, M. R., 1957, The geomagnetic field in Upper Triassic times in the United States: Nature, v. 180, p. 1186–1187.

Dunlop, D. J., and Stirling, J. M., 1977, 'Hard' viscous remanent magnetization (VRM) in fine-grained hematite: Geophysical Research Letters, v. 4, p. 163–166.

Elston, D. P., and Purucker, M. E., 1979, Detrital magnetization in red beds of the Moenkopi Formation (Triassic), Gray Mountain, Arizona: Journal of Geophysical Research, v. 84, p. 1653–1665.

Faust, G. T., 1975, A review and interpretation of the geological setting of the Watchung Basalt flows, New Jersey: United States Geological Survey Professional Paper 864 A, 42 p.

Fitch, E. J., and Miller, J. A., 1971, K-Ar radio ages of Karro volcanic rocks from Lesotho: Bulletin of Volcanology, v. 35, p. 64–84.

Gough, D. I., Brock, A., Jones, D. L., and Opdyke, N. D., 1964, The paleomagnetism of the ring complexes at Marangudzi and the Mateke Hills: Journal of Geophysical Research, v. 69, p. 2489–2497.

Haggerty, S. E., 1976, Oxidation of opaque mineral oxides in basalts, *in* Rumble, D., ed., Oxide minerals: Mineralogical Society of America Short Course Notes, v. 3, p. Hg 1–Hg 100.

Hailwood, E. A., 1975, The paleomagnetism of Triassic and Cretaceous rocks from Morroco: Royal Astronomical Society Geophysical Journal, v. 41, p. 219–235.

Hailwood, E. A., and Mitchell, J. G., 1971, Paleomagnetic and radiometric dating results from Jurassic intrusions in south Morocco: Royal Astronomical Society Geophysical Journal, v. 24, p. 351–364.

Hoblitt, R. P., Larson, E. E., and Walker, T. R., 1974, Chemical remagnetization of Miocene red beds in southern New Mexico [abs.]: EOS (American Geophysical Union Transactions), v. 55, p. 1110.

Hubert, J. F., and Reed, A. A., 1978, Red-bed diagenesis in the East Berlin Formation, Newark Group, Connecticut Valley: Journal of Sedimentary Petrology, v. 48, p. 175–184.

Irving, E., and Banks, M. R., 1961, Paleomagnetic results from Upper Triassic lavas of Massachusetts: Journal of Geophysical Research, v. 66, p. 1935–1939.

Johnson, A. H., 1976, Paleomagnetism of the Jurassic Navajo sandstone from South-western Utah: Royal Astronomical Society Geophysical Journal, v. 44, p. 161–175.

Kelly, W. M., 1979, Magnetic minerals of the red beds of the northern Connecticut Valley—DRM or CRM? [abs.]: EOS (American Geophysical Union Transactions), v. 60, p. 815.

Kluger, K. L., and Sumner, J. R., 1977, Chemical leaching of Triassic red beds from the Newark Basin reveals acquisition of multiple polarities [abs.]: EOS (American Geophysical Union Transactions), v. 58, p. 382.

Kummel, H. B., 1898, The Newark System or red sandstone belt: New Jersey Geological Survey, Annual Report of the State Geologist, 1897, p. 23–159.

Larson, E. E., and Walker, T. R., 1975, Development of chemical magnetization during the early stages of red-bed formation in late Cenozoic sediments, Baja California: Geological Society of America Bulletin, v. 86, p. 639–650.

——1982, A rock-magnetic study of the lower massive sandstone, Moenkopi Formation (Triassic), Gray Mountain Area, Arizona: Journal of Geophysical Research, v. 87, p. 4819–4836.

Larson, E. E., Walker, T. R., Patterson, P. E., Hoblitt, R. P., and Rosenbaum, J., 1982, Paleomagnetism of the Moenkopi Formation, Colorado Plateau: Basis for long-term model of acquisition of chemical remanent magentism in red beds: Journal of Geophysical Research, v. 87, p. 1081–1106.

Manspeizer, W., 1980, Rift tectonics inferred from volcanic and clastic structures, *in* Manspeizer, W., ed., Field studies of New Jersey geology and guide to field trips, 52nd Annual Meeting of the New York State Geological Association: Newark, New Jersey, Rutgers University Press, p. 314–351.

Manspeizer, W., Puffer, J. H., and Cousminer, H. L., 1978, Separation of Morocco and eastern North America: A Triassic-Liassic stratigraphic record: Geological Society of America Bulletin, v. 89, p. 901–920.

Martin, D. L., Nairn, A.E.M., Noltimier, H. C., Petty, M. H., and Schmitt, T. J., 1978, Paleozoic and Mesozoic paleomagnetic results from Morocco: Tectonophysics, v. 44, p. 91–114.

McElhinny, M. W., and Neal, T. S., 1967, Portable field sampling equipment, *in* Collinson, D. W., Creer, K. M., and Runcorn, S. K., eds., Methods in paleomagnetism: Amsterdam, Elsevier, p. 36–40.

McIntosh, W. C., 1976, Paleomagnetic reversals in the Newark Group Brunswick Formation of eastern Pennsylvania and Central New Jersey [Senior thesis]: Princeton, New Jersey, Princeton University, 78 p.

McLaughlin, D. B., 1944, Triassic stratigraphy in the Point Pleasant District, Pennsylvania: Pennsylvania Academy of Science Proceedings, v. 18, p. 62–69.

——1946, The Triassic rocks of the Hunterdon Plateau, New Jersey: Pennsylvania Academy of Science Proceedings, v. 20, p. 89–98.

——1959, Mesozoic rocks, *in* Geology and mineral resources of Bucks County, Pennsylvania: Pennsylvania State Geological Survey Bulletin, v. 9, p. 55–162.

Morel, P., and Irving, E., 1981, Paleomagnetism and the evolution of Pangea: Journal of Geophysical Research, v. 86, p. 1858–1872.

Olsen, P. E., 1978, On the use of the term Newark for Triassic and Early Jurassic rocks of eastern North America: Newsletter of Stratigraphy, v. 7, p. 90–95.

——1980, Triassic and Jurassic Formations in the Newark Basin, *in* Manspeizer, W., ed., Field studies of New Jersey geology and guide to field trips, 52nd Annual Meeting of the New York State Geological Association: Newark, New Jersey, Rutgers University Press, p. 2–41.

Olsen, P. E., McCune, A. R., and Thomson, K. S., 1982, Correlation of the early Mesozoic Newark Supergroup by vertebrates, principally fishes: American Journal of Science, v. 282, p. 1–44.

Opdyke, N. D., 1961, The paleomagnetism of the New Jersey Triassic: A field study of the inclination error in red sediments: Journal of Geophysical Research, v. 66, p. 1941–1949.

Perchevsky, D. M., and Kramov, A. N., 1973, Mesozoic paleomagnetic scale of the USSR: Nature, v. 244, p. 499–501.

Purucker, M. E., Elston, D. P., and Shoemaker, E. M., 1980, Early acquisition of characteristic magnetization in red beds of the Moenkopi Formation (Triassic), Gray Mountain, Arizona: Journal of Geophysical Research, v. 85, p. 997–1012.

Reeve, S. G., and Helsley, C. E., 1972, Magnetic reversal sequence in the upper part of the Chinle Formation, Montoya, New Mexico: Geological Society of America Bulletin, v. 83, p. 3795–3812.

——1974, Magnetization of the Wingate and Chinle Formations, southeastern Utah [abs.]: EOS (American Geophysical Union Transactions), v. 55, p. 1110.

Smith, T. E., 1978, Paleomagnetism of Newark Group arkose, Connecticut and Maryland [abs.]: EOS (American Geophysical Union Transactions), v. 59, p. 271–272.

Smith, T. E., and Noltimier, H. C., 1979, Paleomagnetism of the Newark trend igneous rocks of the north Central Appalachians and the opening of the central Atlantic Ocean: American Journal of Science, v. 279, p. 778–807.

Steiner, M. B., 1978, Magnetic polarity during the Middle Jurassic as recorded in the Sommerville and Curtis formations: Earth and Planetary Science Letters, v. 38, p. 331–345.

Steiner, M. B., and Helsley, C. E., 1974, Magnetic polarity sequence of the Upper Triassic Kayenta Formation: Geology, v. 2, p. 191–199.

Tauxe, L., Kent, D. V., and Opdyke, N. D., 1980, Magnetic components contributing to the NRM of Middle Siwalik red beds: Earth and Planetary Science Letters, v. 47, p. 279–284.

Turner, P., 1979, The paleomagnetic evolution of continental red beds: Geology Magazine, v. 116, p. 289–301.

Van Houton, F. B., 1962, Cyclic sedimentation and the origin of analcime-rich Upper Triassic Lockatong Formation, west-central New Jersey and adjacent Pennsylvania: American Journal of Science, v. 260, p. 561–576.

——1968, Iron oxides in red beds: Geological Society of America Bulletin, v. 83, p. 2761–2772.

——1969, Late Triassic Newark Group, north central New Jersey and adjacent Pennsylvania and New York, *in* Subitzky, S., ed., Geology of selected areas in New Jersey and eastern Pennsylvania: New Brunswick, New Jersey, Rutgers University Press, p. 314–347.

van Zijl, J.S.V., Graham, W. T., and Hales, A. L., 1962, The paleomagnetism of the Stormberg lavas of southern Africa, I: Evidence for a genuine reversal of the Earth's field in Triassic-Jurassic times: Royal Astronomical Society Geophysical Journal, v. 7, p. 23–34.

Walker, T. R., 1967, Formation of red beds in modern and ancient deserts: Geological Society of America Bulletin, v. 78, p. 353–368.

Walker, T. R., Larson, E. E., and Hoblitt, R. P., 1981, The nature and origin of hematite in the Moenkopi Formation (Triassic), Colorado Plateau: A contribution to the origin of magnetism in red beds: Journal of Geophysical Research, v. 86, p. 317–333.

Zijderveld, J.D.A., 1967, A.C. demagnetization of rocks: Analysis of results, *in* Collinson, Creer, and Runcorn, eds., Methods in paleomagnetism, v. 3, p. 254–286.

MANUSCRIPT RECEIVED BY THE SOCIETY NOVEMBER 16, 1982
REVISED MANUSCRIPT RECEIVED JULY 13, 1984
MANUSCRIPT ACCEPTED JULY 27, 1984

Printed in U.S.A.